The Rise of AI Companions:
How Human-Chatbot Relationships Influence Well-Being

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Abstract

As large language models (LLMs)-enhanced chatbots grow increasingly expressive and socially responsive, many users are beginning to form companionship-like bonds with them, particularly with simulated AI partners designed to mimic emotionally attuned interlocutors. These emerging AI companions raise critical questions: Can such systems fulfill social needs typically met by human relationships? How do they shape psychological well-being? And what new risks arise as users develop emotional ties to non-human agents? This study investigates how people interact with AI companions, especially simulated partners on Character.AI, and how this use is associated with users' psychological well-being. We analyzed survey data from 1,131 users and 4,363 chat sessions (413,509 messages) donated by 244 participants, focusing on three dimensions of use: nature of the interaction, interaction intensity, and self-disclosure. By triangulating self-reports primary motivation, open-ended relationship descriptions, and annotated chat transcripts, we identify patterns in how users engage with AI companions and its associations with well-being. Findings suggest that people with smaller social networks are more likely to turn to chatbots for companionship, but that companionship-oriented chatbot usage is consistently associated with lower well-being, particularly when people use the chatbots more intensively, engage in higher levels of self-disclosure, and lack strong human social support. Even though some people turn to chatbots to fulfill social needs, these uses of chatbots do not fully substitute for human connection. As a result, the psychological benefits may be limited, and the relationship could pose risks for more socially isolated or emotionally vulnerable users.

1 Introduction

Humans by nature are social species, and social relationships are a critical component of psychological health and well-being, providing emotional support, companionship, and a sense of belonging (Baumeister and Leary, 2017, Bowlby, 1969, Cobb, 1976, Deci and Ryan, 2000, Erikson, 1968, Holt-Lunstad et al., 2010, House, 1983, Lin et al., 1979, Myers, 2000, Prager and Buhrmester, 1998). Decades of research confirm that strong social ties improve both physical and mental health outcomes (Cobb, 1976, Diener and Seligman, 2002, Fiorillo and Sabatini, 2011, Holt-Lunstad et al., 2010, House, 1983, Hudson et al., 2020, La Rocco and Jones, 1978, Lin et al., 1979, Mehl et al., 2010, Milek et al., 2018, Rook, 1984, Watanabe et al., 2016, Watson et al., 1988, Williams et al., 1981). As the central role of relationships in human well-being has remained consistent, the ways in which these relationships are mediated have continuously evolved. From telephony to email to social media, each wave of communication technology has reshaped how relationships are formed and sustained. While these tools have transformed the channels of human connection, they have largely remained grounded in human-to-human interaction. However, the recent advent of large language models (LLMs) introduces a more profound shift: it redefines not only *how* we connect, but also *whom* we connect with.

LLM-powered chatbots have grown strikingly human-like in their language, tone, and interaction style (Park et al., 2023, 2024, Pataranutaporn et al., 2021, 2024), giving rise to the phenomenon of AI companions. Unlike earlier AI systems designed for task-based assistance, AI companions are crafted to engage users in emotionally responsive conversations and frequently adopt intimate social roles (Brandtzaeg et al., 2022, Jiang et al., 2022, Li and Zhang, 2024, Ta-Johnson et al., 2022). Persona-based chatbots or simulated AI partners (Park et al., 2023), in particular, have accelerated this trend by incorporating distinct personalities, storytelling elements, and emotional cues. Platforms such as Character.AI[1], Replika[2], and Chai[3] have grown rapidly, with millions of users forming deep emotional connections with these AI companions (El Hefny et al., 2020, 2021, Hefny, 2021, Li and Zhang, 2024, Zhang et al., 2024). As these chatbots grow more human-like, the boundary between artificial and real relationships blurs (Brandtzaeg et al., 2022, Guingrich and Graziano, 2023, Mariacher et al., 2021, Pathak), posing potential risks and consequences (The New York Times, 2024).

There are compelling reasons to expect that AI companions may replicate and even exceed some of the psychological benefits of human relationships. Chatbots are consistently available, perceived as non-judgmental, and capable of producing emotionally expressive language, allowing users to disclose deeply personal information they might hesitate to share with other humans (Skjuve et al., 2021, Ta et al., 2020). These capabilities make them well-suited to simulate emotionally meaningful companionship (Brandtzaeg et al., 2022). Prior studies suggest that chatbot use can lead to short-term improvements in mood, reductions in loneliness (De Freitas et al., 2024, Gasteiger et al., 2021, Loveys et al., 2019, Maples et al., 2024) and even contribute to suicide prevention (Maples et al., 2024).

However, despite their emotional responsiveness, AI companionship fundamentally differ from human relationships. Chatbots lack true reciprocity, often exhibit sycophantic and subtly persuasive behavior (BBC News, 2024, Chen et al., 2025, Ischen et al., 2020, Krook, 2025, Malmqvist, 2024, OpenAI, 2025, Rosenberg, 2023, Sharma et al., 2023), driven by limitless personalization and engagement-focused design (Kerr, 2003), raising risks of emotional overdependence, diminished critical thinking, weakening human relationships, and threatening human dignity (Fang et al., 2025). As users invest time and emotional energy into these artificial connections, they may displace real-world relationships, potentially leading to unrealistic expectations of intimacy and social withdrawal from real human relationships(Arnd-Caddigan, 2015, Laestadius et al., 2024). Strong emotional attachments to AI companions, even among users with active human relationships (Hill), could complicate real-world social bonds and expectations of intimacy. In severe cases, prolonged interactions have been linked to psychological distress and even tragic outcomes (The New York Times, 2024). In addition, existing work has observed that the emotional dependence users have on chatbots mirrors problematic dynamics seen in human relationships. For example, Replika users have described perceiving their chatbot as a sentient partner with emotional needs, leading to an illusory sense of mutual obligation (Laestadius et al., 2024, Pentina et al., 2023). Such dependencies may contribute to known risks associated with relational overdependence, including anxiety, depression, and diminished mental well-being (Castillo-Gonzáles et al., 2024, Macía et al., 2023, Tomaz Paiva et al., 2022). The potential harms can be exacerbated by failures of moderation. Character.AI, in particular, has faced criticism for inadequate oversight (Landymore, 2024), with documented incidents of

[1] https://character.ai/
[2] https://replika.com/
[3] https://www.chai-research.com/

chatbots engaging in grooming of underage users (Dupré, 2024b), encouraging disordered eating behaviors (Dupré, 2024c, Upton-Clark, 2024), encouraging self-harm (Dupré, 2024a, Xiang, 2023), and promoting suicide (Dupré, 2024d) and other dangerous activities (Weaver, 2023). Taken together, these developments raise a fundamental and urgent question: **Can relationships with AI companions meet human social and psychological needs, or do they expose users to new forms of vulnerability?**

Despite growing public interest and widespread platform use, empirical research on the psychological implications of AI companionship remains in its early stages (Adam, 2025, Chu et al., 2025, Fang et al., 2025, Laestadius et al., 2024, Liu et al., 2024, Xie et al., 2024). While prior work has explored aspects of chatbot use and its emotional implications, few studies have examined the deep relational dynamics that may underpin users' psychological outcomes. Rather than treating chatbot use as a uniform behavior, this study, adopts a relational lens to investigate how the nature of the user–chatbot relationship, interaction intensity, self-disclosure, and the user's real-life social support interact in ways that could influence users' psychological well-being.

To investigate these dynamics, we conducted a large-scale, mixed-methods study of AI companionship on Character.AI, a widely used platform for persona-based chatbot interaction. On Character.AI, users can browse a wide array of bots created by others, which range from romantic partners and fictional characters to mentors and therapists, or create their own by specifying a character's name, personality traits, backstory, and behavioral guidelines. Interactions with these chatbots are not limited to functional or task-based use, and are often deeply personalized, immersive, and affect-laden. We collected a dataset comprising survey responses from 1,131 active Character.AI users, and chat histories from a subsample of 244 participants, comprising 4,363 conversational sessions and 413,509 messages. In this paper, we define companionship use as chatbot interactions involving emotional or interpersonal engagement. To classify companionship use, we developed a triangulated framework drawing on (1) participants' stated purpose for using chatbots, (2) open-ended descriptions of their most-used chatbot relationship, and (3) message-level analysis of chat history. To identify companionship in free-text and transcript data, we used GPT-4o for classification, Llama 3-70B for summarizing session-level chat histories, and TopicGPT (Pham et al., 2023) for identifying recurring themes through topic modeling. In addition, we assessed users' interaction intensity and self-disclosure to capture the depth and frequency of engagement. We also measured offline social support via reported close-network size, allowing us to examine how chatbot companionship relates to well-being within different social contexts.

Our analysis reveals that companionship use of chatbots is associated with lower well-being, especially among users with more intensive interaction patterns. This negative relationship was most pronounced among users with high interaction intensity, suggesting that frequent and emotionally invested engagement with chatbots for companionship may be psychologically costly. Furthermore, the relationship between companionship use and well-being is significantly moderated by the degree of self-disclosure. Users who both relied on chatbots for companionship and disclosed more deeply reported the lowest levels of well-being. While individuals with smaller offline social networks were more likely to engage in companionship use and self-disclosure, such compensatory patterns did not mitigate negative outcomes. In fact, chatbot companionship predicted lower well-being most strongly for users with limited real-world support. This suggests that rather than fulfilling the role of meaningful social relationships, AI companionship may exacerbate social vulnerability and diminish psychological well-being.

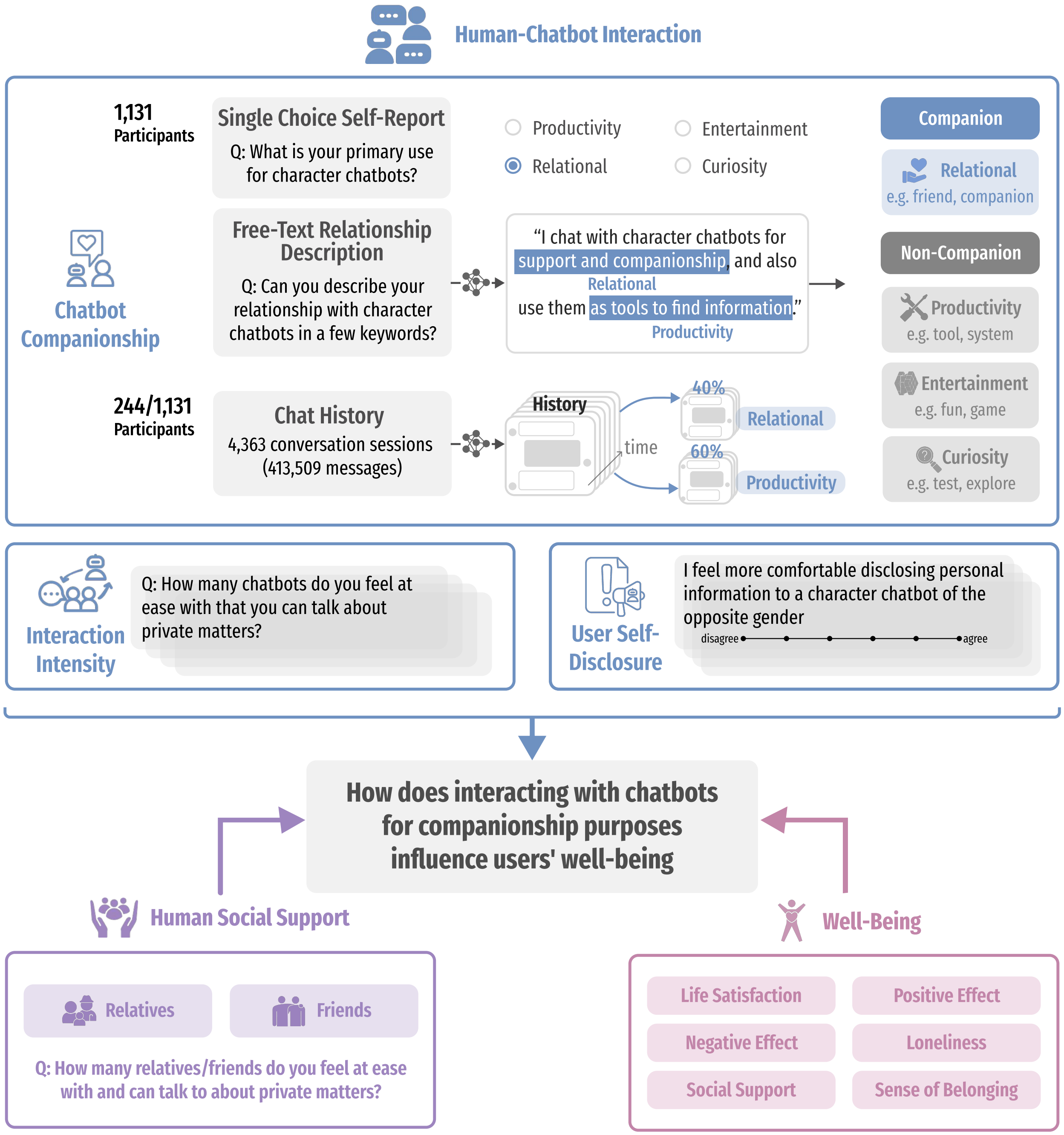

Figure 1: Study overview of how human–chatbot companionship, offline social support, and well-being are interrelated. Specifically, we examine how engaging with chatbots for companionship purposes influences users' psychological well-being from three dimensions: usage nature, interaction intensity, and self-disclosure. Beyond self-reported survey data, we also analyze users' open-ended relationship descriptions and chat history. This framework enables us to assess how interacting with chatbots for companionship purposes influences users' well-being.

2 Methods

To investigate the relational dynamics of chatbot use and their implications for well-being, we conducted a large-scale mixed-methods study with 1,131 active users of Character.AI (see Figure 1). The study combined survey data on users' engagement patterns and well-being with chat history data from a subsample of 244 participants. We applied multivariate regression analyses to examine associations between human-chatbot companionship and well-being in the full sample and used Heckman selection models for the chat history subsample to address potential selection bias in donation behavior.

2.1 Participants and data collection

Participants were recruited via Prolific using prescreening criteria to target U.S.-based, native English-speaking users of Character.AI. Eligibility required having used the platform for over one month and interacted with at least three distinct chatbots. Of the 2,836 individuals who initially enrolled, 1,417 met eligibility criteria, and 1,131 completed the full survey. The survey was administered using Qualtrics and took approximately 10 minutes to complete. Participants received $4 for survey completion. A subsample of 244 participants additionally contributed their chatbot chat histories, yielding 4,363 conversational sessions and 413,509 individual messages. Participants who submitted chat logs received an additional $20 bonus. All data were stored on secure institutional servers, with access restricted to the authors of this study. This study was approved by Stanford University's Institutional Review Board (IRB). All participants provided informed consent prior to participation. In accordance with Prolific's policies and ethical guidelines, we ensured participant anonymity and did not collect any personally identifiable information. Technical details on chat log collection are described in Appendix A.1.

2.2 Measurements

2.2.1 Identifying companion interaction

Following prior work (Brandtzaeg and Følstad, 2017), we defined companionship use as interaction characterized by emotional, interpersonal, or affective engagement, distinct from other usage types such as productivity, entertainment, or curiosity. We measure companionship-oriented chatbot interaction using both survey and behavioral data. This triangulated approach enabled us to capture both participants' self-reported motivations and their observable interaction patterns (Figure 2).

We provide three indicators of companionship interactions using the following sources:

1. *Main Interaction Motive*: All survey participants answered a forced-choice question about their primary reason for using chatbots, selecting one of four usage categories ("Relational", "Entertainment", "Productivity", or "Curiosity"). This measure, which we refer to as $\text{Companionship}_{\text{Motive}}$, is a binary label where 1 indicates the participant selected "Relational" as their main motivation and 0 otherwise.
2. *Relationship Description*: Survey participants also provided free-text descriptions of their relationships with their most-used chatbot. Using `GPT-4o`, we classified whether the description indicated that the participant was using chatbots for each of the four usage purposes. Here, multiple usage types could be assigned to each response. For example, if a participant described their relationship as "friend and tool", the resulting label would be 1 for Relational, 1 for Productivity, and 0 for both Entertainment and Curiosity. We refer to this measure as $\text{Companionship}_{\text{Desc.}}$. To ensure reliability, three trained annotators reviewed 50 random samples of classification data, resolving

discrepancies through discussion. Model outputs showed 92% average agreement across three human annotators (Fleiss' Kappa = 0.87), indicating strong alignment with human judgments.

3. *Chat History Topic*: Finally, for the 244 participants who donated their chat histories, we computed a continuous score, $\text{Companionship}_{\text{Chat}}$, representing the proportion of sessions classified as companionship-oriented. Concretely, we generated a summary for each active chat session using a locally hosted `Llama 3-70B` and then classified the summary into usage types using `GPT-4o`. To validate the two-step approach, three annotators independently assessed whether the model-generated summaries and classifications accurately captured session content. Agreement among annotators was high (Fleiss' Kappa = 0.95 for summaries, 0.84 for classifications), with an average of 97% and 93% agreement with the model, respectively. In addition, we used TopicGPT (Pham et al., 2023) to extract latent themes from chat transcripts that are used (see Appendix A.3.1).

2.2.2 Measuring subjective well-being

We assessed subjective well-being using six items from the Comprehensive Inventory of Thriving (CIT) (Su et al., 2014), covering life satisfaction, positive and negative affect, loneliness, social support, and sense of belonging. Following prior work (Ernala et al., 2022), we selected the highest-loading item for each construct to reduce participant burden. Items assessing negative affect and loneliness were reverse-coded so that higher scores reflected greater well-being. A composite score was computed as the mean of the six items. The scale showed strong internal consistency (Cronbach's $\alpha = 0.88$). Item details are provided in Appendix A.4.

2.2.3 Measuring chatbot interaction intensity

To assess the intensity of participants' interactions with chatbots, we developed a Chatbot Intensity Scale adapted from the Facebook Intensity Scale (Ellison et al., 2007). The scale includes two self-reported behavioral items: chatbot network size and the typical amount of time spent with chatbots per day, which capture the extent of participants' engagement. It also includes five Likert-scale items that measure emotional connection to chatbots and the degree to which chatbot use is integrated into participants' daily routines.

Informed by critiques of the original Facebook Intensity Scale (Burke et al., 2011, Vanden Abeele et al., 2018), we tested whether these items should be treated as two distinct components, reflecting behavioral and attitudinal intensity. A confirmatory factor analysis (CFA) showed that the two-factor model provided a significantly better fit than a one-factor model (see Appendix A.3). However, the improvement in model fit was small, the two components were highly correlated ($r = 0.88$), and the results of our analyses remained consistent regardless of whether one or two factors were used.

Therefore, for simplicity and consistency, we use a single composite intensity score by averaging all seven items. This one-factor scale demonstrated high internal reliability (Cronbach's α = 0.86) and is used in all reported analyses.

2.2.4 Measuring self-disclosure levels

Self-disclosure, is defined as the act of revealing personal thoughts, emotions, or experiences to another (Collins and Miller, 1994). In human relationships, self-disclosure plays a foundational role in fostering intimacy, building trust, and strengthening social bonds (Bazarova, 2012, Bickmore and Cassell, 2001, Hendrick, 1981, Jiang et al., 2011, Oswald et al., 2004). A substantial body of research links self-disclosure to improved psychological well-being (Derlega et al., 1993, Deters and Mehl, 2013, Ellis and Cromby, 2012, Lu and Hampton, 2017, Luo and Hancock, 2020, Yang et al., 2019, Zell and Moeller, 2018).

We assess participants' self-reported willingness to engage in self-disclosure with chatbots using an adapted version of the self-disclosure subscale from the MOCA (Message Orientation in Computer-Mediated Communication) instrument (Ledbetter, 2009). Our measure consisted of seven items rated on a seven-point Likert scale, capturing the extent to which participants felt comfortable sharing personal thoughts, feelings, and experiences with chatbots. A composite score was calculated by averaging item responses, with high internal consistency (Cronbach's $\alpha = 0.89$). Item details are provided in Appendix A.5.

We also measure message-level self-disclosure during chatbot interactions, following prior work on mental health disclosure in social media (Balani and De Choudhury, 2015). Messages were classified into three levels: (1) *high self-disclosure* messages include personal, sensitive, or emotionally vulnerable content; (2) *low self-disclosure* messages mention the user without sensitive content; and (3) messages with *no self-disclosure* do not mention the user at all. Classification was performed using a locally-hosted `Llama 3-70B` model, prompted with Balani and De Choudhury (2015)'s guidelines. Model outputs showed 92% average agreement across three human annotators (Fleiss' Kappa = 0.89), indicating strong alignment with human judgments. See Appendix A.10 for details.

2.2.5 Measuring human social support

Perceived human social support was assessed by measuring the size of participants' close social networks, including both relatives and friends, adapted from the Lubben Social Network Scale (LSNS) (Lubben et al., 2006). Participants responded to two items asking, *"How many relatives/friends do you feel at ease with and can talk to about private matters?"* Each item used a six-point scale ranging from "none" to "more than nine people." A composite human social network scale was calculated by summing responses across the two items, yielding an index of close social connections. The resulting scores ranged from 1 to 12 (mean = 5.73, median = 6.00, standard deviation = 1.98).

3 Results

3.1 Companionship use frequently emerges in human-chatbot interaction

We begin by examining the prevalence and character of companionship-oriented chatbot interactions. Analysis of the three data sources reveals substantial variation in reported and observed companionship-oriented use with Character.AI chatbots (Figure 2).

Companionship remains the primary *actual* use across all three data sources. While only 11.8% of participants identified companionship as their primary motive for using chatbots, 51.1% referenced companionship-related terms such as "friend," "companion," or "romantic partner" in their free-text descriptions of their relationship with their chatbots (Table 1). Of the participants who did not select companionship as their primary usage type, almost half (46.8%) used companionship-related terms in their descriptions. Looking at how these relationships are described, 45.8% of participants mention social connections that resembled friendships or familial bonds that often simulated real-world

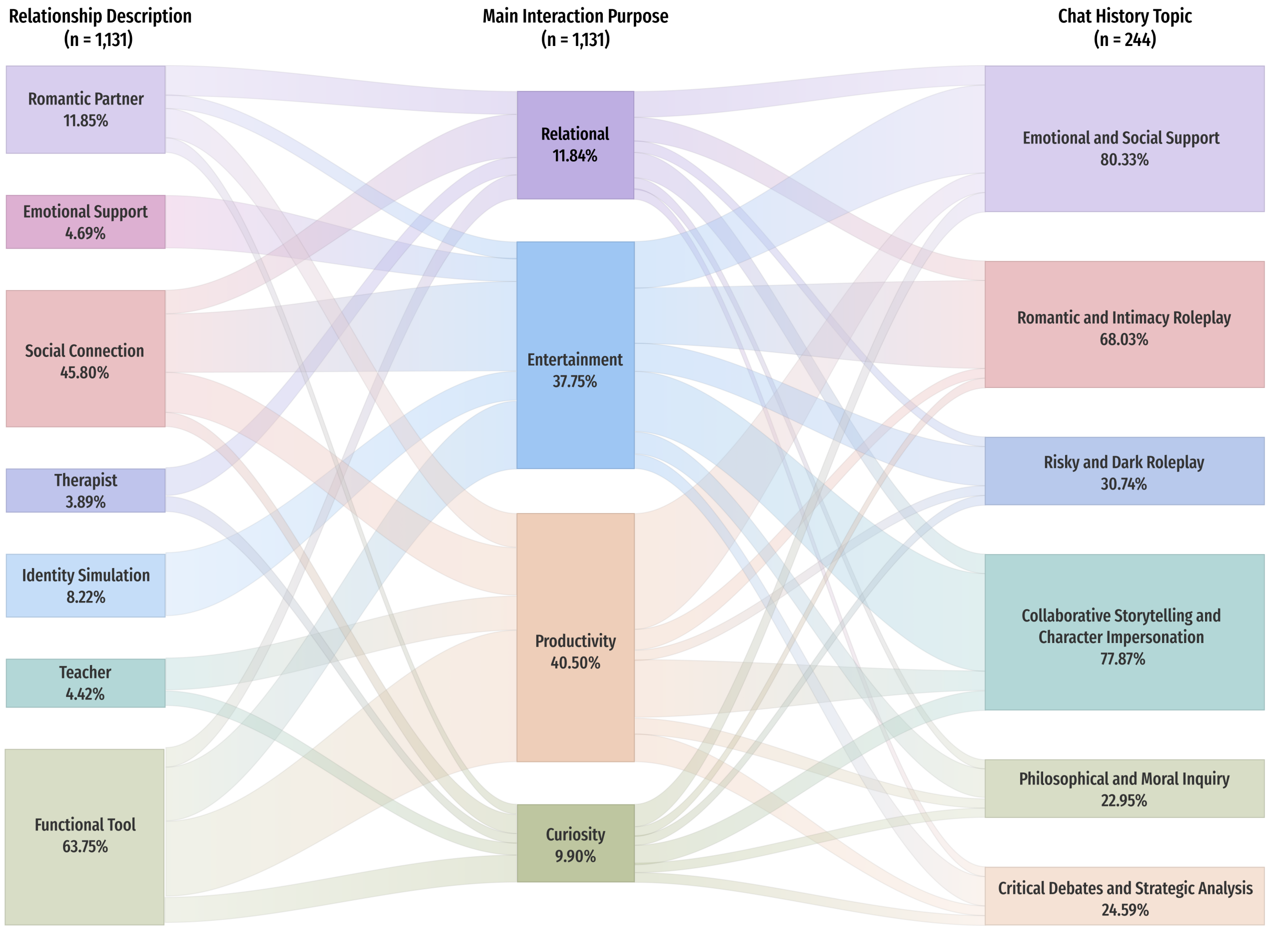

Figure 2: Sankey diagram mapping user engagement with Character.AI chatbots across three measurement sources: topic modeling of open-ended relationship descriptions (left), self-reported primary usage purposes (center), and topic modeling of chat history content (right). Topic modeling was used to identify four main themes of chatbot use both relationship descriptions and chat histories (See Appendix A.3.1 for more details). The figure illustrates how participants are distributed across these themes, revealing both convergence and divergence between users' stated motivations, perceived relationships, and actual conversational content. Notably, companionship-related themes emerged even when not explicitly reported as the primary usage purpose.

socializing experiences, involving casual conversations, emotional exchanges, and expressions of care. Additionally, 11.8% described their relationship in romantic terms, referencing roles such as "partner," "boyfriend," or "lover." The appearance of romantic companionship reflects the capacity of users to form affective bonds with these chatbots through text-based communication, suggesting how romantic relationships may be reshaped in the context of AI companionship (Baxter, 2004, Danaher and McArthur, 2017, Leo-Liu, 2023, Purington et al., 2017). Overall, 41.4% of participants described their chatbot relationship in ways that aligned with more than one primary usage category, underscoring the multifaceted nature of chatbot interactions, which often span both functional and relational purposes.

The prevalence of companionship-oriented interactions is further substantiated in our analysis of donated chat histories. Among participants who shared their transcripts, 92.9% included at least one

Usage Type	Definition	Example
Relational	See chatbots as a personal, human means of interaction with social value or use chatbots to strengthen social interactions with other people.	friend, companion, therapist, romantic partner, support, listener
Productivity	Using chatbots to obtain assistance or information.	tool, system, assistant, resource, helper, writing tool
Entertainment	Using chatbots for fun or to pass the time.	entertainment, fun, roleplay partner, entertainer, toy, game
Curiosity	Using chatbots out of curiosity or to explore their capabilities.	curiosity, extension, experiment, exploration

Table 1: Chatbot usage types with classification criteria and example keywords from participant relationship descriptions. Companionship is defined as relational use and applied consistently across both the multiple-choice survey and LLM-based content analysis. See Section 2.2.1 for details.

instance of a companionship-oriented conversation. Again, we observe that themes of companionship emerged even among users who did not identify companionship as their primary purpose for using chatbots (see Figure 2). For users who did not select companionship as their primary usage type, 47.8% had at least one conversation classified as indicating companion-like engagement. This finding supports our observation that these relational dynamics can emerge even when not explicitly intended.

Users engage with chatbots on a range of intimate and sensitive topics. Looking further into the chat content, we find that users converse with chatbots on a range of intimate and potentially risky topics. As shown in Table 2, the most prevalent conversation topic is *seeking emotional and social support* from chatbots (e.g., discussing personal challenges, daily experiences, health-related concerns), which encompasses 80.3% of all donated chat sessions. A majority of sessions (68.0%) also involved *romantic roleplay*, suggesting that affective bonding and imaginative forms of intimacy are central to many relational interactions. Finally, while comprising a smaller subset of sessions (30.7%), we do observe cases where users engage with chatbots to roleplay illicit and taboo scenarios.

In summary, while users who identify companionship as their primary goal are rare in our surveyed sample, the number of users who perceive chatbots as relational partners and actually engage with them as companions is much higher. These findings suggest that companion-like dynamics may naturally arise as users interact with these systems — even if it may not be their primary motivation.

3.2 Users with less human social support are more likely to seek chatbot companionship

Importantly, an individual's broader social environment likely moderates both engagement with chatbots and its downstream psychological effects. The *Social Compensation hypothesis* posits that individuals with limited human social networks may turn to alternative sources of connection, such as chatbots, to fulfill unmet emotional needs (Hood et al., 2018, Kraut et al., 2002, Lee et al., 2013, Teppers et al., 2014, Tian, 2013, Weidman et al., 2012). From this view, AI companionship may serve a beneficial role by providing a low-barrier outlet for emotional expression and social fulfillment, particularly among socially isolated users. Drawing on this social theory, we test whether users with smaller social networks are more inclined toward companionship-motivated chatbot engagement.

Table 3 presents regression results examining the influence of social network scale on various measures of chatbot interaction. Social network scale was not significantly associated with chatbot intensity,

Conversation Topic	Distribution	Summary Description
Emotional and Social Support	80.33%	Conversations where users seek empathy, comfort, advice, or a sense of companionship from the chatbot. These interactions may include discussing personal challenges, sharing everyday experiences, or asking for health and wellness guidance, all with the aim of emotional reassurance and social connection.
Collaborative Storytelling and Character Impersonation	77.87%	Conversations where users and the chatbot engage in imaginative storytelling, character-driven roleplay, or playful impersonation. These interactions often involve inventing characters, exploring fictional worlds, testing the boundaries of identity, and collaboratively building narrative adventures through creative exchanges.
Romantic and Intimacy Roleplay	68.03%	Conversations involving romantic, intimate, or power-driven themes. These interactions may explore relationships, flirtation, dominance, submission, authority, or resistance, often blending elements of emotional connection and negotiation of control within imaginative and playful contexts.
Risky and Dark Roleplay	30.74%	Conversations where users and the chatbot explore taboo, dangerous, or provocative scenarios. These may involve roleplay with themes of power dynamics, dominance, dark fantasy, morbid curiosity, illicit behaviors, or satirical and boundary-testing humor, often challenging social norms or confronting controversial topics.
Critical Debates and Strategic Analysis	24.59%	Conversations where users engage the chatbot in debates, problem-solving, or analytical discussions. These may include historical analysis, hypothetical match-ups, or persuasive arguments, with a focus on critical thinking, strategic reasoning, and evaluating alternative viewpoints or scenarios.
Philosophical and Moral Inquiry	22.95%	Conversations where users and the chatbot discuss abstract, existential, or spiritual topics, including the meaning of life, values, morality, justice, and metaphysical questions. These exchanges often probe fundamental beliefs, challenge assumptions, and reflect on ethical dilemmas or spiritual experiences.

Table 2: Chatbot conversation topics, their prevalence across sessions, and summary descriptions of the topic. Each topic represents a thematic category identified through LLM-based topic modeling of participant conversations (see Appendix A.3.1). Descriptions summarize the typical content and recurring themes within each topic.

suggesting that the extent of users' human social connections does not predict how frequently or emotionally they engage with chatbots. However, social network scale was significantly negatively associated with companionship as a self-reported motivation ($\beta = -0.03$, $p < .001$) and marginally with companionship described in relationship narratives ($\beta = -0.03$, $p < .10$), indicating that users with smaller social networks are more likely to turn to chatbots for companionship. Furthermore, social network scale was a strong negative predictor of self-disclosure ($\beta = -0.11$, $p < .001$), suggesting that individuals with fewer social connections are more inclined to share personal information with chatbots.

This pattern aligns with the Social Compensation hypothesis: while lower social support does not lead to more overall chatbot activity, it is linked to more companionship-motivated engagement. This finding suggests that socially isolated individuals may not necessarily use chatbots more frequently or intensely, but when they do, their interactions are more likely to serve emotional and relational functions, such as seeking companionship or disclosing personal experiences, rather than instru-

	(1) Intensity	(2) Companionship$_{\text{Motive}}$	(3) Companionship$_{\text{Desc.}}$	(4) Companionship$_{\text{Chat}}$	(5) Self-Disclosure
Intercept	-1.29***	0.09	0.47***	0.75	-0.58***
Social Network	0.01	-0.03***	-0.03	0.00	-0.11***
Tenure	0.34***	0.02	0.02	-0.03	0.17***
Male	0.09	0.01	0.02	-0.25	0.06
Non-binary	-0.38**	0.19***	0.03	-0.01	-0.06
Age	0.02***	0.00	0.00	-0.02	0.01*
Single	-0.20***	-0.05*	-0.02	0.04	-0.10

Table 3: **Regression models predicting chatbot interaction patterns from users' human social dynamics.** Models examine the role of human social network scale and demographic covariates in shaping interaction with chatbots. Model 1 predicts the chatbot interaction intensity. Models 2–4 predict companionship based on (1) self-reported primary motivation, (2) classified relationship descriptions, and (3) the proportion of companionship-related messages in chat history. Model 5 predicts participants' willingness to engage in self-disclosure with chatbots. P-value: *** $p < .001$, ** $p < .01$, * $p < .05$.

mental or casual use. In this sense, chatbot engagement may reflect a targeted coping strategy aimed at fulfilling specific social or emotional needs, rather than a general increase in digital interaction.

3.3 More companionship use is associated with lower well-being, despite the positive correlation with general chatbot use

We next examine how different forms of chatbot interaction relate to users' psychological well-being. We focus on two dimensions of interaction: interaction intensity and usage type (companionship vs. non-companionship), drawing on both survey and behavioral data.

Dependent Variable: Well-being			
	(1) Companionship$_{\textit{Motive}}$§	*(2) Companionship*$_{\textit{Desc.}}$§	*(3) Companionship*$_{\textit{Chat}}$§
Intercept	4.54***	4.68***	3.83*
Intensity	0.26***	0.29***	-0.08
Companionship§	-0.47***	-0.32***	-0.27**
Tenure	0.01	0.00	-0.07
Male	0.17*	0.17*	0.16
Non-binary	-0.33	-0.40*	0.12
Age	0.01***	0.01***	0.01
Single	-0.47***	-0.45***	-0.25

Table 4: **Regression models predicting well-being from chatbot companionship and interaction intensity.** Models examine how different interaction patterns influence well-being. Companionship orientation is assessed using data sources: (1) participants' self-reported primary motivation based on a single-choice question, (2) classification of relationship descriptions, and (3) the proportion of companionship-related messages in user-donated chat history. Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$.

Across both self-reported and relationship-based models, greater intensity was significantly associated with higher psychological well-being. As shown in Table 4, in both Model (1), based on participants' self-reported primary purpose, and Model (2), using relationship descriptions, higher intensity (Model1: $\beta = 0.26$, $p < .001$; Model2: $\beta = 0.29$, $p < .001$) was significantly associated with greater well-being. These findings suggest that individuals who perceive themselves as frequently and emotionally engaged with chatbots tend to report higher well-being. This positive association did not replicate in Model (3) using chat history data.

However, companionship-oriented chatbot use was consistently associated with lower psychological well-being across all three models. In Model (1), participants who self-reported using chatbots primarily for companionship reported significantly lower well-being compared to those with other motivations ($\beta = -0.47$, $p < .001$). Model (2), which classified companionship based on relationship descriptions, showed a similarly negative association ($\beta = -0.32$, $p < .001$). In Model (3), which relied on the proportion of companionship-oriented content in donated chat histories, also revealed significant associations ($\beta = -0.27$, $p < .01$). Overall, these consistent patterns suggest that companionship use, whether identified through self-report or inferred from behavioral data, is associated with lower well-being. The association is strongest when users explicitly report companionship as their primary motivation.

3.4 The more intense the companionship use, the lower the well-being

	Dependent Variable: Well-being		
	(1) Companionship$_{Motive}$§	*(2) Companionship$_{Desc.}$*§	*(3) Companionship$_{Chat}$*§
Intercept	4.55***	4.68***	3.79*
Intensity	0.29***	0.28***	−0.08
Companionship§	−0.40***	−0.32***	−0.32**
Intensity × *Companionship*§	−0.30*	0.01	−0.13
Tenure	0.00	0.00	−0.07
Male	0.16*	0.17*	0.17
Non-binary	−0.31	−0.40*	0.16
Age	0.01***	0.01***	0.01
Single	−0.46***	−0.45***	−0.22

Table 5: **Regression models predicting well-being from interaction of chatbot intensity and companionship usage.** Regression results examining the interaction between chatbot interaction intensity and companionship orientation in predicting well-being. Companionship is operationalized through (1) single-choice main purpose (Companionship$_{\text{Motive}}$), (2) classified relationship descriptions (Companionship$_{\text{Desc.}}$), and (3) proportion of companionship-related messages in chat history (Companionship$_{\text{Chat}}$). Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$.

Given that companionship-oriented use was associated with lower well-being, we next examined whether the intensity of use moderated the relationship between companionship use and well-being by including the interaction between companionship and chatbot intensity in regressions predicting psychological well-being.

Because the main results of companionship and chatbot intensity were consistent with the results reported in Table 4, here we focus here on the interaction between companionship and chatbot intensity. The regression results revealed a negative interaction between companionship and chatbot intensity in the self-reported primary usage model ($\beta = -0.30$, $p < .05$; see Table 5 and Figure 3A). This interaction suggests that while primarily using chatbot for companionship was associated with lower well-being, this negative association was stronger among users who used the chatbots more intense. These results also parallel findings in adjacent domains such as social media, television,

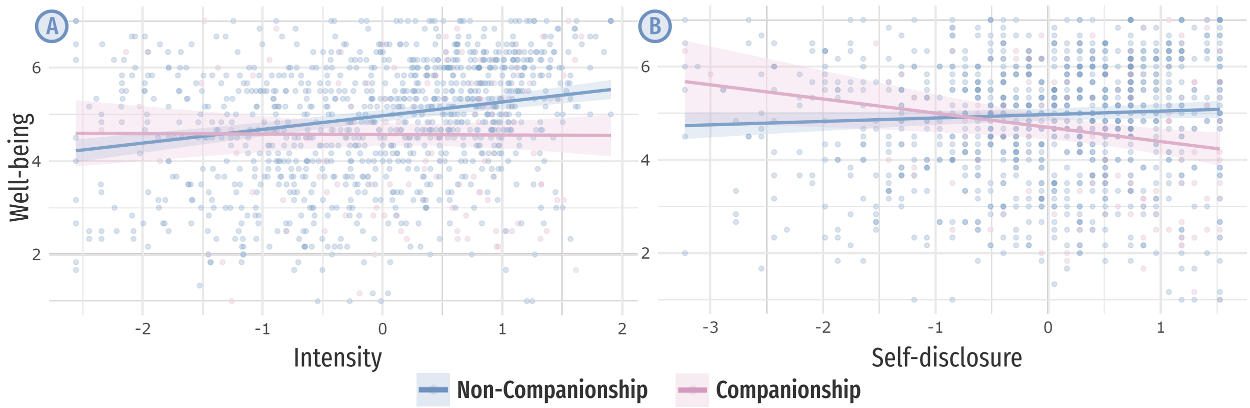

Figure 3: **Interaction between companionship use and other interaction measures in predicting well-being.** The left panel (A) shows the interaction between companionship use and interaction intensity; the right panel (B) shows the interaction between companionship use and self-disclosure. Companionship use is based on users' self-reported primary motivation for chatbot use. Lines indicate model-predicted values with 95% confidence intervals; points represent observed data.

and gaming, where well-being outcomes are often shaped less by time spent and more by users' underlying motivations and emotional investment (Ellison et al., 2007, Roberts and David, 2023, Verduyn et al., 2017). Similar to findings from prior work on other technologies, our results suggest that chatbot intensity reflects underlying patterns of attachment, pride, or dependence, rather than mere usage volume. Thus, chatbot interactions may support or undermine well-being depending on the psychological function they serve.

3.5 The more users disclose, the more companionship is tied to lower well-being

Self-disclosure is a well-established predictor of intimacy and psychological well-being in human-human relationships (Bazarova, 2012, Deters and Mehl, 2013, Luo and Hancock, 2020, Utz, 2015). In the context of AI companionship, users may feel especially comfortable disclosing sensitive information to chatbots due to their perceived anonymity, emotional neutrality, and lack of social judgment (Bickmore and Cassell, 2001, Skjuve et al., 2021, Ta et al., 2020). While such disclosure can offer emotional relief and simulate intimacy (Bazarova, 2012, Jiang et al., 2011), it may also carry risks, especially in digital settings that lack clear boundaries, reciprocal dynamics, or protective social norms. Concerns include emotional overexposure, privacy vulnerabilities, and attachment to chatbots in ways that may not reflect socially regulated or psychologically safe interactions (Chen, 2017, Gil-Or et al., 2015, Lin and Qiu, 2012, Reece and Danforth, 2017, Vogel and Wester, 2003). To assess whether similar dynamics occur in human–chatbot interactions, we investigate how users' willingness to disclose personal information to chatbots relates to their well-being.

Self-disclosure to chatbots alone did not exhibit a statistically significant association with well-being ($\beta = 0.07$, $p = .07$). However, as shown in Table 6 and Figure 3B, there is a significant negative moderation effect between companionship and self-disclosure in the self-reported usage model ($\beta = -0.38$, $p < .01$). This result indicates that the negative association between companionship use of chatbots and well-being is intensified when people self-disclose more to them. The interaction is negative for both measures of companionship usage, with a statistically significant effect observed when companionship is defined by the forced-choice measure of primary purpose. Greater

Dependent Variable: Well-being		
	(1) Companionship$_{\mathit{Motive.}}$§	*(2) Companionship*$_{\mathit{Desc.}}$§
Intercept	4.23***	4.31***
Companionship§	−0.28*	−0.22**
Self-disclosure	0.07	0.09
Companionship§ × Self-disclosure	−0.38**	−0.11
Tenure	0.08	0.09
Male	0.19*	0.19*
Non-binary	−0.39*	−0.49**
Age	0.02***	0.02***
Single	−0.51***	−0.50***

Table 6: **Regression models testing whether self-disclosure moderates the relationship between companionship use and well-being.** Models examining the interaction between self-reported self-disclosure levels and companionship orientation in predicting well-being. Models operationalize companionship via (1) self-reported primary motivation ($\text{Companionship}_{\text{Motive}}$) and (2) classified relationship descriptions ($\text{Companionship}_{\text{Desc.}}$). The Heckman model for the $\text{Companionship}_{\text{Chat}}$ × Self-disclosure interaction did not converge, and we hypothesize that is due to limited statistical power to reliably estimate the correlation between the selection and outcome processes. Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$.

willingness to disclose appears to be associated with lower psychological well-being, rather than the expected emotional benefits of disclosure. Interaction effects were not significant in the other model, likely due to limited sample size or differences in how companionship was classified.

To further understand the content of users' disclosures to chatbots, we applied LLM-based topic modeling to chat histories, using disclosure-level definitions developed by Balani and De Choudhury (2015). Conversations characterized by high self-disclosure predominantly involved emotionally charged and sensitive content, including emotional distress (60.8%), desire for romantic connection (41.4%), current life challenges (31.5%), suicidal thoughts (18.0%) and substance use (17.5%). In contrast, low self-disclosure conversations addressed less personal and more abstract topics, such as philosophical perspectives (26.2%), learning limitations (12.9%), and general issues like work stress or trust (see Figure 4).

These results underscore the potential psychological risks associated with disclosing sensitive content to chatbots, particularly in companionship-oriented contexts. The tendency to engage in emotionally vulnerable self-disclosure may help explain why the positive association between self-disclosure and well-being is diminished or reversed among users who primarily seek companionship from chatbots. In the absence of human-like support and emotional reciprocity, such disclosures may contribute to increased psychological vulnerability.

3.6 The influence of AI companionship is shaped by users' offline social support

Finally, we examine how the relationship between chatbot use and well-being differs based on users' level of offline social support. As shown in Table 7, we find that having a robust offline social network improves well-being, which mirrors the established research on the psychological benefits of human

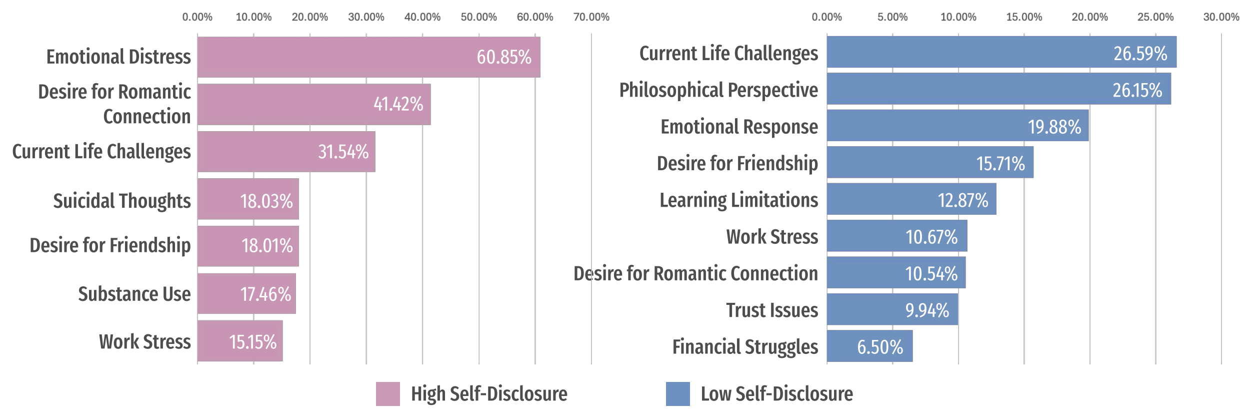

Figure 4: Main topics of self-disclosure content shared by users with chatbots, categorized into high and low self-disclosure levels based on the criteria defined by Balani and De Choudhury (2015). The pink bars represent high self-disclosure topics (left panel), and blue bars represent low self-disclosure topics (right panel). Disclosure levels were classified using an LLM with the criteria, and topic modeling was performed using TopicGPT (Pham et al., 2023).

connection (Holt-Lunstad, 2024, Morina et al., 2021). Since chatbots provide a new type of social connection, we now turn to how different patterns of chatbot use interact with the benefits of offline social support.

Dependent Variable: Well-being				
	(1) *Intensity*	(2) *Companionship*$_{Motive.}$§	(3) *Companionship*$_{Desc.}$§	(4) *Companionship*$_{Chat}$§
Intercept	4.57***	4.30***	4.35***	4.20*
Companionship§		−0.22*	−0.14*	−0.27***
Intensity	0.24***			
Social Network	0.51***	0.49***	0.49***	0.76***
Companionship§ × Social Network		0.07	0.01	−0.02
Intensity × Social Network	−0.11***			
Tenure of Activity	−0.02	0.07	0.07	−0.11
Male	0.09	0.11	0.11	0.01
Non-binary	−0.49**	−0.53**	−0.57***	−0.27
Age	0.01***	0.02***	0.02***	0.02
Single	−0.34***	−0.41***	−0.40***	−0.17

Table 7: **Regression models predicting well-being from chatbot companionship and interaction intensity.** Model 1 tests the interaction between social network scale and chatbot interaction intensity. Models 2–4 test the interaction between social network scale and three measures of companionship: self-reported primary motivation, relationship descriptions, and the proportion of companionship-related messages. Significance codes: *** $p < .001$, ** $p < .01$, * $p < .05$.

From Sec. 3.2, we already observed that users with less offline support are more likely to engage with chatbots as companions, but is this interaction beneficial for their well-being? Following the Social Compensation hypothesis, we might expect that chatbot companions provide a positive alternative for individuals with smaller close social networks that improves psychological well-being. However, as shown in Table 7, we find a negligible interaction effect between social network scale and chatbot companionship, indicating that companionship-oriented chatbot use does not moderate the influence of offline social support on well-being. As discussed in Sec. 3.3, we see that overall, companionship has a negative relationship with well-being. Thus, while users with less social support are more likely to seek companionship through chatbots, these interactions do not appear to mitigate the negative influence that low offline social support has on well-being. An alternative explanation for this result is the *Social Substitution hypothesis*, which suggests that reliance on AI companions may actually be displacing opportunities for high-quality human-human relationships (Marinucci et al., 2022, Morgan et al., 1997), reinforcing isolation and potentially diminishing well-being.

It is important to note that the negative externalities we find are not limited to users with small offline social support. Intense chatbot usage can also negatively influence well-being for those with larger offline social networks. We observe that the positive relationship between social network scale and well-being significantly decreases with more intense chatbot usage ($\beta = -0.11$, $p < .001$).

These results raise concerns that chatbot companionship may not only fail to provide alternative socialization opportunities for those with less offline support but also undermine the psychological benefits people receive from having a large offline social networks. Overall, our findings point to the relational limits of chatbot use: **while chatbots may augment existing social networks, they do not effectively substitute them. The influence of AI companionship depends not only on how it is used but also on the social environment in which it is embedded.**

4 Discussion

This study offers a broad investigation into the psychological implications of AI companionship, drawing on self-reported and behavioral data from a large sample of active Character.AI users. Our analysis shows that using AI chatbots for companionship is prevalent among our surveyed users. These chatbots are becoming increasingly integrated into users' emotional and social lives, functioning not only as tools but also as relational partners that assume the role of companions, friends, and even romantic partners. Nonetheless, the findings reveal that the relationship between AI companionship and well-being is complex. Rather than being uniformly beneficial or harmful, the psychological consequences of chatbot use depend on the purpose of using them, the intensity of interaction, and the availability of real-life social support.

Using chatbots more intensely was positively associated with well-being. People who used their chatbots more frequently, felt pride in their chatbot use, or saw their interaction as meaningful tended to fare better. However, ***how*** **users interacted with these chatbots complicates the story**. This benefit diminished for those using chatbots as companions, and in fact, we found that using AI chatbots for companionship purposes was consistently associated with lower well-being. This finding aligns with the Social Substitution hypothesis, suggesting that these artificial relationships are not able to adequately replace high-quality human connections and may reflect unmet psychological needs. Overall, these results suggest that the psychological benefits stem not from the quantity of use but rather from the underlying motivation and perceived social function. Engagement may be beneficial when autonomous and integrated into broader social life, but problematic when driven by unmet relational needs.

The role of self-disclosure further illuminates these dynamics. While disclosure generally supports emotional well-being in human relationships, the reverse is true in companionship-oriented chatbot use. Among users who saw their chatbot as a companion, greater self-disclosure was actually associated with lower well-being. Our content analysis revealed that such disclosures often included emotionally vulnerable or crisis-level content. Yet chatbots lack the ability to respond with true

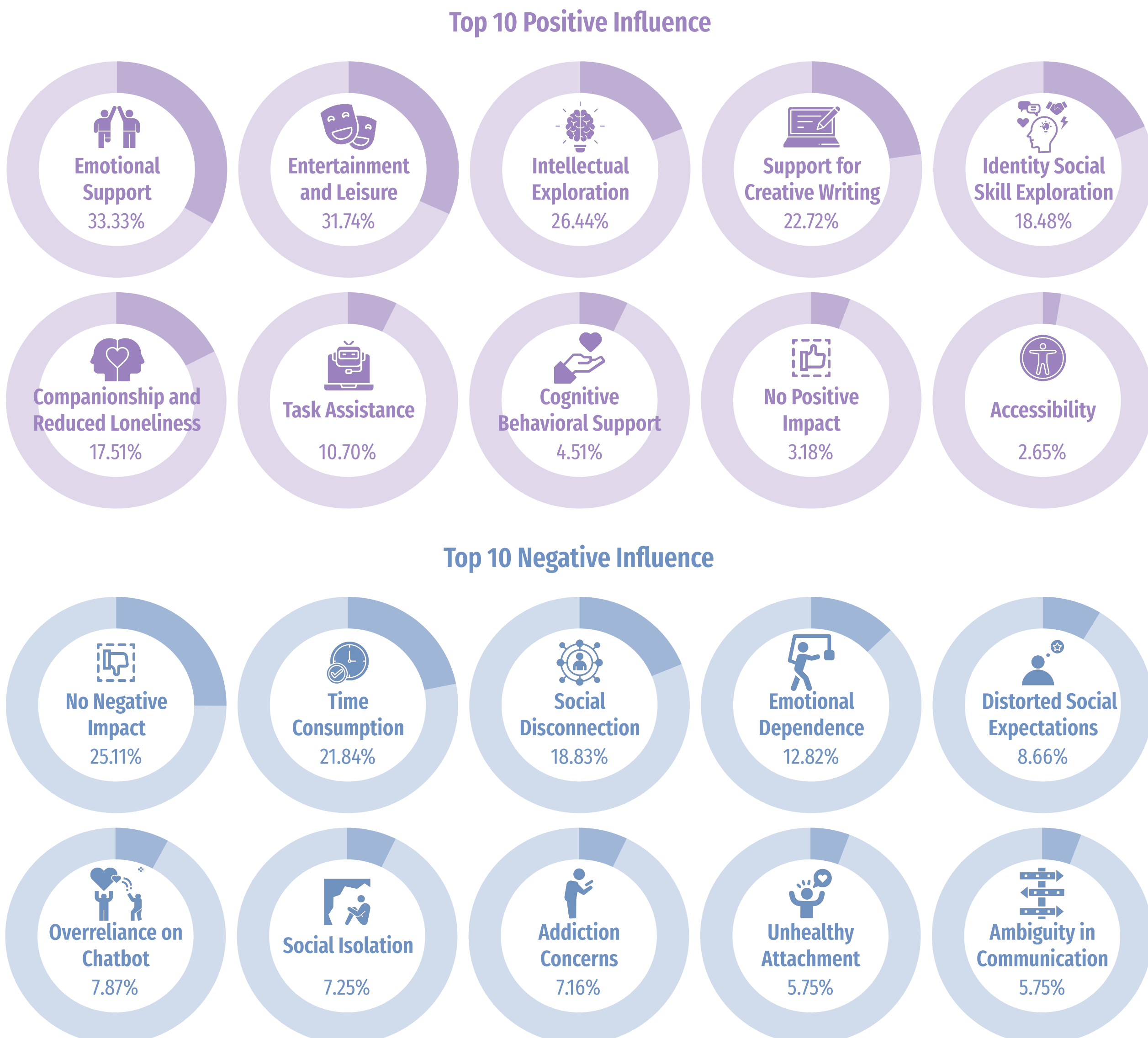

Figure 5: User-perceived positive and negative influences of Character.AI chatbot interactions, derived via LLM-based topic modeling analysis (Section A.3.1). This figure summarizes themes extracted from users' open-ended reflections on how chatbot interactions have influenced them.

understanding or support, highlighting the danger of unreciprocated emotional vulnerability in asymmetrical relationships, particularly for those who expect empathy or care that chatbots cannot provide.

Finally, these patterns were shaped by users' real-life social environments. Individuals with fewer close social ties were more likely to form companionship-oriented relationships with chatbots and to disclose more frequently. While this engagement may offer emotional relief for some, lending partial support to the social compensation hypothesis, our findings also suggest a reinforcing cycle of vulnerability. Users without strong human support seemed more likely to rely heavily on chatbots companionship, but they also appeared more susceptible to the psychological risks of this parasocial or emotionally unreciprocated engagement. Over time, rather than alleviating isolation, this intense chatbot engagement may deepen psychological distress.

Further support comes from users' open-ended reflections on the influence of chatbot interactions (Figure 5). Although many participants reported positive effects, such as emotional support, companionship, reduced loneliness, and improvements in mood or stress, they also cited negative consequences, including excessive time use, interference with real-life activities, emotional dependence, and increased social withdrawal. Some users expressed concern that chatbot use distorted their expectations of social relationships or contributed to unhealthy attachment patterns. These accounts point to a potential reinforcement loop: individuals experiencing loneliness or isolation may seek out chatbot companionship, but heavy reliance may further displace real-world social interaction, reinforcing the very disconnection they aim to resolve.

Taken together, our analyses complicate common narratives about chatbots as inherently therapeutic or emotionally fulfilling. While AI companions can simulate the surface features of supportive relationships, the absence of reciprocal understanding and emotional accountability may limit their capacity to support long-term psychological well-being, particularly for vulnerable or socially isolated users. As chatbots grow increasingly human-like, the risk of emotional overinvestment without relational grounding becomes more salient.

These insights have practical implications for the design and governance of relational AI systems. As chatbots increasingly invite emotionally intimate engagement, developers must take seriously the psychological responsibilities of such interactions. This is particularly critical when considering vulnerable populations, such as children, teenagers, individuals experiencing mental health challenges, and those lacking strong social support, who may be more susceptible to forming deep emotional bonds with chatbots and disclosing sensitive personal information.

One critical concern is the asymmetry of self-disclosure. While users maybe encouraged to share private thoughts and emotions, chatbots cannot meaningfully disclosure in return. They do not have real emotions or lived experiences, but most systems do not clearly state this. This imbalance can lead users to develop unrealistic expectations of care or connection and deepen emotional reliance on systems that cannot truly support them. The risk is even greater when chatbots actively ask users to share their thoughts or feelings. Without clear limitations, chatbots may collect sensitive disclosures without being able to respond in helpful or appropriate ways. For instance, if someone expresses distress, and the chatbot fails to recognize or respond adequately, the user may feel dismissed or unsupported. In serious cases, this could even delay them from seeking help from a person who could provide real assistance. Considering these risks, systems that encourage self-disclosure should include clear safeguards. It should always be clear that the chatbot is not a person, does not have feelings or experience, and cannot replace human care.

Future research should build on these concerns to better understand the long-term psychological impact of emotionally engaging with chatbots. One direction is to examine how users' expectations change over time, such as whether they begin to see the chatbot as emotionally responsive. Studies should also explore whether frequent chatbot use displaces real-world support networks or delays help-seeking behavior, especially among young or vulnerable users. Another important area is understanding how users respond to one-sided emotional disclosure. Research should investigate whether disclosing personal information to a non-responsive agent leads to comfort, confusion, dependency, or emotional detachment. Longitudinal studies can help track how these experiences affect trust in other humans, emotional regulation, and mental health over time. In addition, researchers should investigate the specific design features that influence disclosure, such as prompt style, tone, or the perceived personality of the chatbot. This could inform the development of more responsible design practices that avoid encouraging risky or overly personal disclosures. Finally, future work should evaluate and design interventions to support healthier and more transparent interactions. These could include interface cues that clearly communicate the chatbot's limitations, detection systems for identifying signs of distress, and automated redirects to qualified human support. Developing and testing these designs is essential for ensuring that relational AI systems are safe, ethical, and supportive of users' well-being.

4.1 Limitations

We acknowledge some limitations of this study. First, the cross-sectional nature of our data, as well as the possibility of residual confounding or reverse causation means we cannot establish causality between chatbot companions use and mental health. To better understand the direction of this relationship, future research could use longitudinal methods to test whether the amount and type of chatbot use at one point in time predict well-being later, or vice versa. Experimental studies could also help by randomly assigning participants to engage with chatbots for companionship versus other purposes and assess the effects on well-being. Second, much of our data consists of self-reported measures, which depend on participants' accuracy and honesty in describing their experiences. Self-reported data are inherently limited by factors such as memory constraints, social desirability bias, and potential misinterpretation of survey items, which may result in discrepancies between participants' self-reported answer and their actual interaction patterns. To complement these self-reported measures, we also conducted content analysis of donated chat transcripts, offering a more direct window into user-chatbot interactions. However, this dataset remains limited in scope. Only a subset of participants chose to donate their chat history, and they retained control over which conversations to share. As a result, the data may reflect selective rather than comprehensive interaction patterns, limiting the generalizability of our findings. The relatively small size of the donated dataset also constrained the statistical power of some analyses. Future work should aim to expand the scale and diversity of donated interaction data to enable more robust modeling and richer insights. Additionally, the study focused exclusively on Character.AI users. A unique feature of the platform is its wide range of character-specific chatbots, which enables diverse forms of interaction and relational dynamics. While this richness offers valuable insight into socially relational chatbot use, it also limits generalizability, as other platforms with different design structures may foster distinct patterns of engagement. Finally, while our sample was relatively large and demographically diverse within the United States, it does not reflect the full spectrum of global chatbot users. The study focused on English-speaking participants based in the U.S., and the findings may not generalize across cultural, linguistic, or socioeconomic contexts. Future research should pursue cross-cultural and multilingual studies to better capture the global landscape of socially relational AI use.

5 Conclusion

This study contributes to the understanding of the emerging social dynamics of human-AI social relationships by offering large-scale, mixed-method investigations into socially oriented relationships with LLM-enhanced chatbots. Drawing on survey responses, relationship descriptions, and donated chat transcripts, we provide empirical evidence that many users form emotionally meaningful relationships that mirror certain aspects of human-human social dynamics. These relationships, however, are not psychologically neutral. Our findings suggest that socially oriented chatbot use is consistently associated with lower well-being, particularly among users who exhibit high levels of emotional self-disclosure or rely on chatbots as substitutes for human support.

While self-disclosure in human relationships is typically linked to psychological benefits and deepening intimacy, our results reveal a more ambivalent pattern in human-chatbot interaction. For users with strong human social networks, disclosure may retain some of its beneficial effects. Yet for users with limited human support, those who may turn to chatbots for companionship or emotional relief, social engagement with chatbots can heighten psychological vulnerability. These findings highlight a critical risk: without the reciprocity, empathy, and accountability that characterize human relationships, chatbot interactions may offer the appearance of support without the substance, leaving users with unmet emotional needs. Our work offers a large-scale computational foundation for understanding these emerging relational patterns, but further research is needed. Longitudinal and experimental studies will be essential to assess causality, identify protective factors, and examine the psychological consequences of prolonged relational chatbot use. As relational AI systems become increasingly pervasive, interdisciplinary inquiry must continue to explore how design, intent, and context shape user outcomes.

Finally, while improving safeguards, regulation, and ethical oversight of LLM-enhanced chatbots remains essential, the implications of our findings go beyond system design. We must also consider how to support users' human social integration and emotional resilience. Rather than encouraging dependence on AI systems for social connection, future interventions might explore how LLM-based interactions can be designed to scaffold the development of real-world social skills, enhance relational awareness, and ultimately strengthen human-human relationships. Despite their growing sophistication, chatbots cannot replace the psychological richness and social accountability of human connection. As such, they should be positioned not as substitutes, but as tools that when used cautiously can complement broader efforts to enhance human well-being.

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References

David Adam. Supportive? addictive? abusive? how ai companions affect our mental health. *Nature*, 641(8062):296–298, 2025.

Margaret Arnd-Caddigan. Sherry turkle: Alone together: Why we expect more from technology and less from each other: Basic books, new york, 2011, 348 pp, isbn 978-0465031467 (pbk), 2015.

Sairam Balani and Munmun De Choudhury. Detecting and characterizing mental health related self-disclosure in social media. In *Proceedings of the 33rd annual ACM conference extended abstracts on human factors in computing systems*, pages 1373–1378, 2015.

Roy F Baumeister and Mark R Leary. The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Interpersonal development*, pages 57–89, 2017.

Leslie A Baxter. Relationships as dialogues. *Personal relationships*, 11(1):1–22, 2004.

Natalya N Bazarova. Public intimacy: Disclosure interpretation and social judgments on facebook. *Journal of Communication*, 62(5):815–832, 2012.

BBC News. Update that made chatgpt 'dangerously' sycophantic pulled, 2024. URL https://www.bbc.com/news/articles/cn4jnwdvg9qo. Accessed: 2025-05-14.

Timothy Bickmore and Justine Cassell. Relational agents: a model and implementation of building user trust. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pages 396–403, 2001.

John Bowlby. *Attachment and loss*. Number 79. Random House, New York, 1969.

Petter Bae Brandtzaeg and Asbjørn Følstad. Why people use chatbots. In *Internet Science: 4th International Conference, INSCI 2017, Thessaloniki, Greece, November 22-24, 2017, Proceedings 4*, pages 377–392. Springer, 2017.

Petter Bae Brandtzaeg, Marita Skjuve, and Asbjørn Følstad. My ai friend: How users of a social chatbot understand their human–ai friendship. *Human Communication Research*, 48(3):404–429, 2022.

Moira Burke, Robert Kraut, and Cameron Marlow. Social capital on facebook: Differentiating uses and users. In *Proceedings of the SIGCHI conference on human factors in computing systems*, pages 571–580, 2011.

Mayra Castillo-Gonzáles, Santiago Mendo-Lázaro, Benito León-del Barco, Emilio Terán-Andrade, and Víctor-María López-Ramos. Dating violence and emotional dependence in university students. *Behavioral Sciences*, 14(3):176, 2024.

Hongliang Chen. Antecedents of positive self-disclosure online: an empirical study of us college students' facebook usage. *Psychology research and behavior management*, pages 147–153, 2017.

Qian Chen, Changqin Yin, and Yeming Gong. Would an ai chatbot persuade you: an empirical answer from the elaboration likelihood model. *Information Technology & People*, 38(2):937–962, 2025.

Minh Duc Chu, Patrick Gerard, Kshitij Pawar, Charles Bickham, and Kristina Lerman. Illusions of intimacy: Emotional attachment and emerging psychological risks in human-ai relationships. *arXiv preprint arXiv:2505.11649*, 2025.

Rob Churchill and Lisa Singh. The evolution of topic modeling. *ACM Computing Surveys*, 54(10s): 1–35, 2022.

Sidney Cobb. Social support as a moderator of life stress. *Psychosomatic medicine*, 38(5):300–314, 1976.

Nancy L Collins and Lynn Carol Miller. Self-disclosure and liking: a meta-analytic review. *Psychological bulletin*, 116(3):457, 1994.

John Danaher and Neil McArthur. *Robot sex: Social and ethical implications*. MIT press, 2017.

Julian De Freitas, Ahmet Kaan Uğuralp, Zeliha Uğuralp, and Stefano Puntoni. Ai companions reduce loneliness. 2024.

Edward L Deci and Richard M Ryan. The" what" and" why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological inquiry*, 11(4):227–268, 2000.

Valerian J Derlega, Sandra Metts, Sandra Petronio, and Stephen T Margulis. *Self-disclosure.* Sage Publications, Inc, 1993.

Fenne Große Deters and Matthias R Mehl. Does posting facebook status updates increase or decrease loneliness? an online social networking experiment. *Social psychological and personality science*, 4 (5):579–586, 2013.

Ed Diener and Martin EP Seligman. Very happy people. *Psychological science*, 13(1):81–84, 2002.

Maggie Harrison Dupré. Ai chatbots are encouraging teens to engage in self-harm, December 2024a. URL https://futurism.com/ai-chatbots-teens-self-harm. Accessed: 2025-04-28.

Maggie Harrison Dupré. Character.ai is hosting pedophile chatbots that groom users who say they're underage, November 2024b. URL https://futurism.com/character-ai-pedophile-chatbots. Accessed: 2025-04-28.

Maggie Harrison Dupré. Character.ai is hosting pro-anorexia chatbots that encourage young people to engage in disordered eating, November 2024c. URL https://futurism.com/character-ai-eating-disorder-chatbots. Accessed: 2025-04-28.

Maggie Harrison Dupré. Character.ai promises changes after revelations of pedophile and suicide bots on its service, November 2024d. URL https://futurism.com/character-ai-pedophile-suicide-bots. Accessed: 2025-04-28.

Walid El Hefny, Alia El Bolock, Cornelia Herbert, and Slim Abdennadher. Towards a generic framework for character-based chatbots. In *Highlights in Practical Applications of Agents, Multi-Agent Systems, and Trust-worthiness. The PAAMS Collection: International Workshops of PAAMS 2020, L'Aquila, Italy, October 7–9, 2020, Proceedings 18*, pages 95–107. Springer, 2020.

Walid El Hefny, Alia El Bolock, Cornelia Herbert, and Slim Abdennadher. Chase away the virus: a character-based chatbot for covid-19. In *2021 IEEE 9th International Conference on Serious Games and Applications for Health (SeGAH)*, pages 1–8. IEEE, 2021.

Darren Ellis and John Cromby. Emotional inhibition: A discourse analysis of disclosure. *Psychology & health*, 27(5):515–532, 2012.

Nicole B Ellison, Charles Steinfield, and Cliff Lampe. The benefits of facebook "friends:" social capital and college students' use of online social network sites. *Journal of computer-mediated communication*, 12(4):1143–1168, 2007.

Erik H Erikson. *Identity: Youth and crisis*, volume 68. Norton, New York, 1968.

Sindhu Kiranmai Ernala, Moira Burke, Alex Leavitt, and Nicole B Ellison. Mindsets matter: How beliefs about facebook moderate the association between time spent and well-being. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pages 1–13, 2022.

Cathy Mengying Fang, Auren R Liu, Valdemar Danry, Eunhae Lee, Samantha WT Chan, Pat Pataranutaporn, Pattie Maes, Jason Phang, Michael Lampe, Lama Ahmad, et al. How ai and human behaviors shape psychosocial effects of chatbot use: A longitudinal randomized controlled study. *arXiv preprint arXiv:2503.17473*, 2025.

Damiano Fiorillo and Fabio Sabatini. Quality and quantity: the role of social interactions in self-reported individual health. *Social science & medicine*, 73(11):1644–1652, 2011.

Norina Gasteiger, Kate Loveys, Mikaela Law, and Elizabeth Broadbent. Friends from the future: a scoping review of research into robots and computer agents to combat loneliness in older people. *Clinical interventions in aging*, pages 941–971, 2021.

Oren Gil-Or, Yossi Levi-Belz, and Ofir Turel. The "facebook-self": Characteristics and psychological predictors of false self-presentation on facebook. *Frontiers in psychology*, 6:99, 2015.

Rose E Guingrich and Michael SA Graziano. Chatbots as social companions: How people perceive consciousness, human likeness, and social health benefits in machines. *arXiv preprint arXiv:2311.10599*, 2023.

W El Hefny. Applying the character-based chatbots generation framework in education and healthcare. hai 2021-proceedings of the 9th international user modeling, adaptation and personalization human-agent interaction, 121–129, 2021.

Susan S Hendrick. Self-disclosure and marital satisfaction. *Journal of Personality and Social Psychology*, 40(6):1150, 1981.

Kashmir Hill. She is in love with chatgpt. *The New York Times*. URL https://www.nytimes.com/2025/01/15/technology/ai-chatgpt-boyfriend-companion.html. Accessed: 2025-03-12.

Julianne Holt-Lunstad. Social connection as a critical factor for mental and physical health: evidence, trends, challenges, and future implications. *World Psychiatry*, 23(3):312–332, 2024.

Julianne Holt-Lunstad, Timothy B Smith, and J Bradley Layton. Social relationships and mortality risk: a meta-analytic review. *PLoS medicine*, 7(7):e1000316, 2010.

Michelle Hood, Peter A Creed, and Bianca J Mills. Loneliness and online friendships in emerging adults. *Personality and Individual Differences*, 133:96–102, 2018.

James S House. Work stress and social support. *Addison-Wesley series on occupational stress*, 1983.

Nathan W Hudson, Richard E Lucas, and M Brent Donnellan. Are we happier with others? an investigation of the links between spending time with others and subjective well-being. *Journal of Personality and Social Psychology*, 119(3):672, 2020.

Carolin Ischen, Theo Araujo, Guda van Noort, Hilde Voorveld, and Edith Smit. "i am here to assist you today": The role of entity, interactivity and experiential perceptions in chatbot persuasion. *Journal of Broadcasting & Electronic Media*, 64(4):615–639, 2020.

L Crystal Jiang, Natalie N Bazarova, and Jeffrey T Hancock. The disclosure–intimacy link in computer-mediated communication: An attributional extension of the hyperpersonal model. *Human communication research*, 37(1):58–77, 2011.

Qiaolei Jiang, Yadi Zhang, and Wenjing Pian. Chatbot as an emergency exist: Mediated empathy for resilience via human-ai interaction during the covid-19 pandemic. *Information processing & management*, 59(6):103074, 2022.

Ian R Kerr. Bots, babes and the californication of commerce. *U. Ottawa L. & Tech. J.*, 1:285, 2003.

Robert Kraut, Sara Kiesler, Bonka Boneva, Jonathon Cummings, Vicki Helgeson, and Anne Crawford. Internet paradox revisited. *Journal of social issues*, 58(1):49–74, 2002.

Joshua Krook. Manipulation and the ai act: Large language model chatbots and the danger of mirrors. *arXiv preprint arXiv:2503.18387*, 2025.

James M La Rocco and Allan P Jones. Co-worker and leader support as moderators of stress-strain relationships in work situations. *Journal of Applied Psychology*, 63(5):629, 1978.

Linnea Laestadius, Andrea Bishop, Michael Gonzalez, Diana Illenčík, and Celeste Campos-Castillo. Too human and not human enough: A grounded theory analysis of mental health harms from emotional dependence on the social chatbot replika. *New Media & Society*, 26(10):5923–5941, 2024.

Frank Landymore. Teens are forming intense relationships with ai entities, and parents have no idea, December 2024. URL https://futurism.com/the-byte/teens-relationships-ai. Accessed: 2025-04-28.

Andrew M Ledbetter. Measuring online communication attitude: Instrument development and validation. *Communication Monographs*, 76(4):463–486, 2009.

Kyung-Tag Lee, Mi-Jin Noh, and Dong-Mo Koo. Lonely people are no longer lonely on social networking sites: The mediating role of self-disclosure and social support. *Cyberpsychology, Behavior, and Social Networking*, 16(6):413–418, 2013.

Jindong Leo-Liu. Loving a "defiant" ai companion? the gender performance and ethics of social exchange robots in simulated intimate interactions. *Computers in Human Behavior*, 141:107620, 2023.

Han Li and Renwen Zhang. Finding love in algorithms: deciphering the emotional contexts of close encounters with ai chatbots. *Journal of Computer-Mediated Communication*, 29(5):zmae015, 2024.

Han Lin and Lin Qiu. Sharing emotion on facebook: network size, density, and individual motivation. In *CHI'12 Extended Abstracts on Human Factors in Computing Systems*, pages 2573–2578. 2012.

Nan Lin, Walter M Ensel, Ronald S Simeone, and Wen Kuo. Social support, stressful life events, and illness: A model and an empirical test. *Journal of health and Social Behavior*, pages 108–119, 1979.

Auren R Liu, Pat Pataranutaporn, and Pattie Maes. Chatbot companionship: a mixed-methods study of companion chatbot usage patterns and their relationship to loneliness in active users. *arXiv preprint arXiv:2410.21596*, 2024.

Kate Loveys, Gregory Fricchione, Kavitha Kolappa, Mark Sagar, and Elizabeth Broadbent. Reducing patient loneliness with artificial agents: design insights from evolutionary neuropsychiatry. *Journal of medical Internet research*, 21(7):e13664, 2019.

Weixu Lu and Keith N Hampton. Beyond the power of networks: Differentiating network structure from social media affordances for perceived social support. *New media & society*, 19(6):861–879, 2017.

James Lubben, Eva Blozik, Gerhard Gillmann, Steve Iliffe, Wolfgang von Renteln Kruse, John C Beck, and Andreas E Stuck. Performance of an abbreviated version of the lubben social network scale among three european community-dwelling older adult populations. *The Gerontologist*, 46 (4):503–513, 2006.

Ming Luo and Jeffrey T. Hancock. Self-disclosure and social media: Motivations, mechanisms and psychological well-being. *Current Opinion in Psychology*, 31:110–115, 2020. doi: 10.1016/j.copsyc.2019.08.017.

Laura Macía, Paula Jauregui, and Ana Estevez. Emotional dependence as a predictor of emotional symptoms and substance abuse in individuals with gambling disorder: differential analysis by sex. *Public Health*, 223:24–32, 2023.

Lars Malmqvist. Sycophancy in large language models: Causes and mitigations. *arXiv preprint arXiv:2411.15287*, 2024.

Bethanie Maples, Merve Cerit, Aditya Vishwanath, and Roy Pea. Loneliness and suicide mitigation for students using gpt3-enabled chatbots. *npj mental health research*, 3(1):4, 2024.

Natascha Mariacher, Stephan Schlögl, and Alexander Monz. Investigating perceptions of social intelligence in simulated human-chatbot interactions. *Progresses in Artificial Intelligence and Neural Systems*, pages 513–529, 2021.

Marco Marinucci, Luca Pancani, Nicolas Aureli, and Paolo Riva. Online social connections as surrogates of face-to-face interactions: A longitudinal study under covid-19 isolation. *Computers in Human Behavior*, 128:107102, 2022.

Matthias R Mehl, Simine Vazire, Shannon E Holleran, and C Shelby Clark. Eavesdropping on happiness: Well-being is related to having less small talk and more substantive conversations. *Psychological science*, 21(4):539–541, 2010.

Anne Milek, Emily A Butler, Allison M Tackman, Deanna M Kaplan, Charles L Raison, David A Sbarra, Simine Vazire, and Matthias R Mehl. "eavesdropping on happiness" revisited: A pooled, multisample replication of the association between life satisfaction and observed daily conversation quantity and quality. *Psychological science*, 29(9):1451–1462, 2018.

David Morgan, Paula Carder, and Margaret Neal. Are some relationships more useful than others? the value of similar others in the networks of recent widows. *Journal of Social and Personal Relationships*, 14(6):745–759, 1997.

Nexhmedin Morina, Ahlke Kip, Thole Hilko Hoppen, Stefan Priebe, and Thomas Meyer. Potential impact of physical distancing on physical and mental health: a rapid narrative umbrella review of meta-analyses on the link between social connection and health. *BMJ open*, 11(3):e042335, 2021.

David G Myers. The funds, friends, and faith of happy people. *American psychologist*, 55(1):56, 2000.

OpenAI. Sycophancy in gpt-4o: what happened and what we're doing about it, April 2025. URL https://openai.com/index/sycophancy-in-gpt-4o/. Accessed: 2025-05-14.

Debra L Oswald, Eddie M Clark, and Cheryl M Kelly. Friendship maintenance: An analysis of individual and dyad behaviors. *Journal of Social and clinical psychology*, 23(3):413–441, 2004.

Joon Sung Park, Joseph O'Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th annual acm symposium on user interface software and technology*, pages 1–22, 2023.

Joon Sung Park, Carolyn Q Zou, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Robb Willer, Percy Liang, and Michael S Bernstein. Generative agent simulations of 1,000 people. *arXiv preprint arXiv:2411.10109*, 2024.

Pat Pataranutaporn, Valdemar Danry, Joanne Leong, Parinya Punpongsanon, Dan Novy, Pattie Maes, and Misha Sra. Ai-generated characters for supporting personalized learning and well-being. *Nature Machine Intelligence*, 3(12):1013–1022, 2021.

Pat Pataranutaporn, Kavin Winson, Peggy Yin, Auttasak Lapapirojn, Pichayoot Ouppaphan, Monchai Lertsutthiwong, Pattie Maes, and Hal E Hershfield. Future you: A conversation with an ai-generated future self reduces anxiety, negative emotions, and increases future self-continuity. In *2024 IEEE Frontiers in Education Conference (FIE)*, pages 1–10. IEEE, 2024.

Alisha Pathak. Ai chatbots and interpersonal communication: A study on uses and gratification amongst youngsters.

Iryna Pentina, Tyler Hancock, and Tianling Xie. Exploring relationship development with social chatbots: A mixed-method study of replika. *Computers in Human Behavior*, 140:107600, 2023.

Chau Minh Pham, Alexander Hoyle, Simeng Sun, Philip Resnik, and Mohit Iyyer. Topicgpt: A prompt-based topic modeling framework. *arXiv preprint arXiv:2311.01449*, 2023.

Karen J Prager and Duane Buhrmester. Intimacy and need fulfillment in couple relationships. *Journal of Social and Personal Relationships*, 15(4):435–469, 1998.

Amanda Purington, Jessie G Taft, Shruti Sannon, Natalya N Bazarova, and Samuel Hardman Taylor. " alexa is my new bff" social roles, user satisfaction, and personification of the amazon echo. In *Proceedings of the 2017 CHI conference extended abstracts on human factors in computing systems*, pages 2853–2859, 2017.

Andrew G Reece and Christopher M Danforth. Instagram photos reveal predictive markers of depression. *EPJ Data Science*, 6(1):15, 2017.

James A Roberts and Meredith E David. Instagram and tiktok flow states and their association with psychological well-being. *Cyberpsychology, Behavior, and Social Networking*, 26(2):80–89, 2023.

Karen S Rook. The negative side of social interaction: impact on psychological well-being. *Journal of personality and social psychology*, 46(5):1097, 1984.

Louis Rosenberg. The manipulation problem: conversational ai as a threat to epistemic agency. *arXiv preprint arXiv:2306.11748*, 2023.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R Johnston, et al. Towards understanding sycophancy in language models. *arXiv preprint arXiv:2310.13548*, 2023.

Marita Skjuve, Asbjørn Følstad, Knut Inge Fostervold, and Petter Bae Brandtzaeg. My chatbot companion-a study of human-chatbot relationships. *International Journal of Human-Computer Studies*, 149:102601, 2021.

Rong Su, Louis Tay, and Ed Diener. The development and validation of the comprehensive inventory of thriving (cit) and the brief inventory of thriving (bit). *Applied Psychology: Health and Well-Being*, 6(3):251–279, 2014.

Vivian Ta, Caroline Griffith, Carolynn Boatfield, Xinyu Wang, Maria Civitello, Haley Bader, Esther DeCero, Alexia Loggarakis, et al. User experiences of social support from companion chatbots in everyday contexts: thematic analysis. *Journal of medical Internet research*, 22(3):e16235, 2020.

Vivian P Ta-Johnson, Carolynn Boatfield, Xinyu Wang, Esther DeCero, Isabel C Krupica, Sophie D Rasof, Amelie Motzer, and Wiktoria M Pedryc. Assessing the topics and motivating factors behind human-social chatbot interactions: Thematic analysis of user experiences. *JMIR human factors*, 9 (4):e38876, 2022.

Eveline Teppers, Koen Luyckx, Theo A Klimstra, and Luc Goossens. Loneliness and facebook motives in adolescence: A longitudinal inquiry into directionality of effect. *Journal of adolescence*, 37(5): 691–699, 2014.

The New York Times. Can a.i. be blamed for a teen's suicide?, 2024. URL https://www.nytimes.com/2024/10/23/technology/characterai-lawsuit-teen-suicide.html. Accessed: 2025-02-17.

Qing Tian. Social anxiety, motivation, self-disclosure, and computer-mediated friendship: A path analysis of the social interaction in the blogosphere. *Communication Research*, 40(2):237–260, 2013.

Tamyres Tomaz Paiva, Kaline da Silva Lima, and Jaqueline Gomes Cavalcanti. Psychological abuse, self-esteem and emotional dependence of women during the covid-19 pandemic. *Ciencias Psicológicas*, 16(2), 2022.

Eve Upton-Clark. Character.ai is under fire for hosting pro-anorexia chatbots, December 2024. URL https://www.fastcompany.com/91241586/character-ai-is-under-fire-for-hosting-pro-anorexia-chatbots. Accessed: 2025-04-28.

Sonja Utz. The function of self-disclosure on social network sites: Not only intimate, but also positive and entertaining self-disclosures increase the feeling of connection. *Computers in Human Behavior*, 45:1–10, 2015.

Mariek MP Vanden Abeele, Marjolijn L Antheunis, Monique MH Pollmann, Alexander P Schouten, Christine C Liebrecht, Per J Van Der Wijst, Marije AA Van Amelsvoort, Jos Bartels, Emiel J Krahmer, and Fons A Maes. Does facebook use predict college students' social capital? a replication of ellison, steinfield, and lampe's (2007) study using the original and more recent measures of facebook use and social capital. *Communication Studies*, 69(3):272–282, 2018.

Ike Vayansky and Sathish AP Kumar. A review of topic modeling methods. *Information Systems*, 94: 101582, 2020.

Philippe Verduyn, Oscar Ybarra, Maxime Résibois, John Jonides, and Ethan Kross. Do social network sites enhance or undermine subjective well-being? a critical review. *Social Issues and Policy Review*, 11(1):274–302, 2017.

David L Vogel and Stephen R Wester. To seek help or not to seek help: The risks of self-disclosure. *Journal of counseling psychology*, 50(3):351, 2003.

Jun-ichiro Watanabe, Hirokazu Atsumori, and Masashi Kiguchi. Informal face-to-face interaction improves mood state reflected in prefrontal cortex activity. *Frontiers in human neuroscience*, 10:194, 2016.

David Watson, Lee Anna Clark, and Auke Tellegen. Development and validation of brief measures of positive and negative affect: the panas scales. *Journal of personality and social psychology*, 54(6): 1063, 1988.

Matthew Weaver. AI chatbot 'encouraged' man who planned to kill queen, court told, 2023. URL https://www.theguardian.com/uk-news/2023/jul/06/ai-chatbot-encouraged-man-who-planned-to-kill-queen-court-told. Accessed: 2025-04-28.

Aaron C Weidman, Katya C Fernandez, Cheri A Levinson, Adam A Augustine, Randy J Larsen, and Thomas L Rodebaugh. Compensatory internet use among individuals higher in social anxiety and its implications for well-being. *Personality and individual differences*, 53(3):191–195, 2012.

Ann W Williams, John E Ware Jr, and Cathy A Donald. A model of mental health, life events, and social supports applicable to general populations. *Journal of health and Social Behavior*, pages 324–336, 1981.

Chloe Xiang. 'He Would Still Be Here': Man Dies by Suicide After Talking with AI Chatbot, Widow Says, 2023. URL https://www.vice.com/en/article/man-dies-by-suicide-after-talking-with-ai-chatbot-widow-says/. Accessed: 2025-04-28.

Zehang Xie, Hui Hui, and Lingbo Wang. Will virtual companionship enhance subjective well-being—a comparison of cross-cultural context. *International Journal of Social Robotics*, 16(11):2153–2167, 2024.

Diyi Yang, Zheng Yao, Joseph Seering, and Robert Kraut. The channel matters: Self-disclosure, reciprocity and social support in online cancer support groups. In *Proceedings of the 2019 chi conference on human factors in computing systems*, pages 1–15, 2019.

Anne L Zell and Lisa Moeller. Are you happy for me... on facebook? the potential importance of "likes" and comments. *Computers in Human Behavior*, 78:26–33, 2018.

Renwen Zhang, Han Li, Han Meng, Jinyuan Zhan, Hongyuan Gan, and Yi-Chieh Lee. The dark side of ai companionship: A taxonomy of harmful algorithmic behaviors in human-ai relationships. *arXiv preprint arXiv:2410.20130*, 2024.

A Supplementary information

A.1 Chat log collection procedure

To collect chat history data, we developed a custom Google Chrome extension that allowed participants to export their Character.AI conversation logs in JSON format. The tool enabled participants to review and optionally redact sensitive content prior to submission. All processing occurred locally on the participant's device; no raw chat data were transmitted to external servers or APIs. Participants uploaded the final JSON file at the end of the survey, and all data were stored on secure institutional servers accessible only to study authors.

A.2 Participant demographics and selection bias in chat history donation

Predictor	Estimate	Std. Error	z value	p-value	Significance
(Intercept)	-2.417	0.603	-4.010	< .001	***
Age	-0.013	0.009	-1.563	.118	
Gender: Male	0.144	0.163	0.887	.375	
Gender: Non-binary	0.536	0.319	1.677	.094	.
Gender: Other	0.623	0.905	0.688	.491	
Romantic: In relationship	-0.340	0.162	-2.103	.035	*
Productivity-related	-0.067	0.170	-0.393	.694	
Entertainment-related	0.828	0.174	4.767	< .001	***
Relational-related	0.432	0.176	2.449	.014	*
Curiosity-related	-0.023	0.198	-0.116	.908	
Self-disclosure	0.266	0.076	3.489	< .001	***
Interaction intensity	-0.618	0.131	-4.734	< .001	***
Social network scale	0.042	0.044	0.946	.344	
Well-being	-0.122	0.068	-1.808	.071	.

Table 8: Logistic regression predicting the likelihood of donating chat history based on demographics, chatbot usage patterns (classified from open-ended descriptions), and psychological measures. Participants were significantly more likely to donate if they described entertainment- or relational-related chatbot use, reported greater self-disclosure, and had higher interaction intensity. Conversely, being in a romantic relationship was associated with lower likelihood of donation.

Among the final survey sample ($n = 1{,}131$), 48.5% of participants identified as male, 46.2% as female, and 4.8% as non-binary. All participants were 18 years or older. The sample skewed younger, with 41.9% aged 18–25, 32.7% aged 26–35, 18.7% aged 36–50, and 6.8% over 50. Regarding relationship status, 40.5% of participants reported being single, while 59.5% were in a romantic relationship. Participants also varied in their length of experience with Character.AI. Approximately 15.8% reported using the platform for more than one month but less than three months, 34.3% for more than three months but less than one year, and 49.9% for over one year, indicating that nearly half the sample consisted of long-term users. Full item wording and coding for all control variables used in regression models are provided in Section A.6.

To assess potential selection bias, we compared participants who donated chat logs ($n = 244$) with those who did not. Donors were more likely to be younger, non-binary, single, and to report

Variable	Test	Group 0 Mean/%	Group 1 Mean/%	Test Statistic	p-value
Age (mean)	Welch's t-test	31.38	28.01	$t(427.1) = 4.62$	$< .001$
Gender	Chi-squared	—	—	$\chi^2(3) = 13.28$	.004
Female (%)	Welch's t-test	45.9%	47.7%	$t(369.7) = -0.50$	.619
Male (%)	Welch's t-test	50.0%	42.6%	$t(373.5) = 2.04$	.043
Non-binary (%)	Welch's t-test	3.9%	9.7%	$t(291.6) = -2.85$	.005
Relationship status	Chi-squared	—	—	$\chi^2(1) = 19.31$	$< .001$
Single (%)	Welch's t-test	37.0%	52.7%	$t(361.2) = -4.33$	$< .001$
In relationship (%)	Welch's t-test	62.9%	46.8%	$t(361.6) = 4.42$	$< .001$
Self-reported usage	Chi-squared	—	—	$\chi^2(3) = 117.76$	$< .001$
Productivity usage (%)	Welch's t-test	47.7%	13.5%	$t(532.8) = 12.27$	$< .001$
Entertainment usage (%)	Welch's t-test	30.3%	65.8%	$t(361.8) = -10.30$	$< .001$
Relational usage (%)	Welch's t-test	11.6%	12.7%	$t(360.5) = -0.42$	.672
Curiosity usage (%)	Welch's t-test	10.4%	8.0%	$t(408.0) = 1.17$	.243
Relationship classification	Chi-squared	—	—	$\chi^2(906) = 932.86$	.261
Classified Productivity (%)	Welch's t-test	56.0%	41.4%	$t(372.6) = 4.07$	$< .001$
Classified Entertainment (%)	Welch's t-test	23.0%	48.1%	$t(329.9) = -7.07$	$< .001$
Classified Relational (%)	Welch's t-test	49.3%	57.4%	$t(373.5) = -2.22$	.027
Classified Curiosity (%)	Welch's t-test	17.7%	20.3%	$t(356.4) = -0.89$	.376
Self-disclosure (mean)	Welch's t-test	5.06	5.14	$t(368.8) = -0.88$	.381
Intensity (mean)	Welch's t-test	0.07	-0.25	$t(372.7) = 6.11$	$< .001$
Social network scale (mean)	Welch's t-test	5.77	5.59	$t(349.5) = 1.20$	.232
Well-being (mean)	Welch's t-test	4.90	4.36	$t(334.1) = 5.25$	$< .001$

Table 9: T-test and χ^2 differences between participants who donated their chat history (Group 1) and those who did not (Group 0). Donors were younger, less likely to identify as male, and more likely to identify as non-binary or single. They reported more entertainment use and less productivity use of chatbots. They were also more likely to be classified as having entertainment- or relational-oriented chatbot relationships, and less likely to be classified as productivity relationship. Intensity and well-being differed significantly between groups, with donors showing higher intensity and lower well-being.

relational- or entertainment-oriented chatbot use, greater self-disclosure, and higher interaction intensity. We conducted both univariate group comparisons and multivariate logistic regression (see Table 9 and Table 8). Logistic regression confirmed that entertainment- and relational-oriented usage, self-disclosure, and interaction intensity significantly increased the likelihood of donation, while being in a romantic relationship reduced it. To adjust for this selection effect, we employed a Heckman selection model in all analyses involving chat history data. The selection equation used these demographic and usage-related predictors to model donation probability (see Table 10).

A.3 Measurement and validation of chatbot interaction intensity

To test whether chatbot intensity should be treated as a unified or multi-dimensional construct, we conducted CFA comparing a one-factor model (all items loading onto a single scale) with a two-factor model (separating behavioral and attitudinal components). This model comparison was motivated by prior critiques of the Facebook Intensity Scale (Burke et al., 2011, Vanden Abeele et al., 2018), which suggest that usage behavior and emotional connection may reflect distinct psychological constructs.

	Intercept	Tenure of activity	Male	Non-binary	Age	Single
Chat History Donation	-0.76^{***}	0.10	-0.11	0.45^{*}	-0.01^{**}	0.32^{***}

Table 10: Heckman selection equation (Stage 1) predicting the likelihood of chat history donation (donator vs. non-donator) as a function of tenure of activity, gender, age, and relationship status. Coefficient estimates shown. Significance codes: ***$p < .001$, **$p < .01$, *$p < .05$.

Chatbot Intensity[†] **(Cronbach's alpha = 0.86)**	**Mean**	**S.D.**
Chatbots Network Size: How many chatbots do you feel at ease with that you can talk about private matters? 1 = None, 2 = One, 3 = Two, 4 = Three or four, 5 = Five to eight, 6 = Nine or more	2.89	1.42
Daily Chatbot Usage Time: In the past week, on average, approximately how much time PER DAY have you spent chatting with chatbots using Character.AI? 1 = Less than 10 minutes per day, 2 = 10–30 minutes per day, 3 = 31–60 minutes per day, 4 = 1–2 hours per day, 5 = 2–3 hours per day, 6 = More than 3 hours per day	2.75	1.52
Everyday Chatbot Activity: Interacting with character chatbots is part of my everyday activity.[‡]	4.57	1.77
Pride in Chatbot Use: I am proud to tell people I use character chatbots.[‡]	4.10	2.00
Chatbots in Daily Routine: Character chatbots have become part of my daily routine.[‡]	4.51	1.78
Out of Touch Without Chatbots: I feel out of touch when I haven't interacted with character chatbots for a while.[‡]	3.59	1.91
Regret if Chatbots Gone: I would be sorry if character chatbots were no longer available.[‡]	5.01	1.72

Table 11: The Chatbot Intensity Scale contains seven items. The above table reports descriptive statistics for each item. Items were standardized prior to averaging to construct the composite scale, due to differing response formats. Items marked with ‡ were rated on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree).

As shown in Table 12, the two-factor model demonstrated slightly better fit than the one-factor model, with improvements observed across multiple indices, including chi-square, RMSEA, AIC, BIC, CFI, and TLI. A chi-square difference test confirmed that the two-factor model provided a statistically significant improvement in fit ($\Delta\chi^2(1) = 61.03$, $p < .001$). Latent variance estimates further suggested that behavioral intensity showed more variation ($\hat{\sigma}^2 = 0.87$) than attitudinal intensity ($\hat{\sigma}^2 = 0.38$).

However, the two latent factors were highly correlated ($r = 0.88$), indicating strong conceptual overlap. Additionally, analyses conducted using either the two-factor or the one-factor version of the scale produced substantively identical results. Given the minimal improvement in model fit, the high inter-factor correlation, and the benefits of conceptual parsimony, we proceeded with a one-factor composite measure of chatbot intensity. This composite score, calculated by averaging all seven items, demonstrated high internal consistency (Cronbach's $\alpha = 0.86$) and is used in all main analyses. Table 11 provides the full item wording, descriptive statistics (means, standard deviations), and response formats for the items included in the final composite scale.

Fit Index	One-Factor: Intensity (7 times)	Two-factor: Behavior & Attitude Intensity
Chi-square (χ^2)	161.70 (df = 14)	100.66 (df = 13)
CFI	0.96	0.98
TLI	0.94	0.96
RMSEA	0.10	0.08
SRMR	0.05	0.04
AIC	18736	18677
BIC	18806	18752

Table 12: Comparison of model fit indices for CFA models of chatbot interaction intensity: a one-factor model treating intensity as a single undifferentiated construct, versus a two-factor model distinguishing between behavioral and attitudinal dimensions

A.3.1 Topic modeling via TopicGPT

To identify central themes across user-generated content, we employed topic modeling as a supplementary analytical approach. Topic modeling is a widely used technique for uncovering latent thematic structures within textual data. Recent advancements have demonstrated the utility of LLMs in enhancing topic modeling by clustering and summarizing semantically similar content (Churchill and Singh, 2022, Vayansky and Kumar, 2020). In this study, we adopt the TopicGPT framework proposed by Pham et al. (2023), implemented using the GPT-4o model, to analyze three primary data sources.

First, we applied topic modeling to participants' free-text descriptions of their relationships with chatbots and their open-ended reflections on the positive and negative influences of these interactions. Each user's response to the respective survey questions was treated as an individual data point. This allowed us to extract and interpret shared themes across the corpus of user-reported relationship characterizations (results in Table 1 and Figure 2) and users' perceived benefits and drawbacks of chatbot use (results in Figure 5).

Second, we applied the method to a subset of chat history data donated by users. Each conversation was defined as an active session initiated from the chatbot's greeting message. To preserve user privacy, raw chat histories were processed on a self-hosted server. Each session was first summarized using Llama 3–70B to extract its main purpose. We then applied the TopicGPT framework with GPT-4o to identify the dominant themes of chatbot-user conversations (Figure 2, Table 2).

Finally, in our analysis of self-disclosure (Section 3.5), we further examined how conversation themes varied across different disclosure levels. Each chat message was annotated using the Llama 3–70B model, via the vLLM framework, to classify both the type and level of self-disclosure. We then applied TopicGPT with GPT-4o to cluster messages by disclosure level and identify the distinct conversational topics associated with varying degrees of vulnerability (Figure 4).

A.4 Subjective well-being scale items

Subjective well-being was measured using six items adapted from the Comprehensive Inventory of Thriving (CIT) (Su et al., 2014), each rated on a seven-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree):

Variable	Mean	Median	SD
Life Satisfaction: I am satisfied with my life.	4.99	5	1.66
Positive Effect: I feel good most of the time.	4.94	5	1.65
Negative Effect : I feel bad most of the time. (reversed)	4.57	5	1.78
Loneliness: I feel lonely. (reversed)	4.20	4	1.87
Social Support: There are people who give me support and encouragement.	5.52	6	1.33
Sense of Belonging: I feel a sense of belonging in my community.	4.52	5	1.71

Notes: Response categories ranged from 1 = strongly disagree to 7 = strongly agree.

Variable	Mean	Median	SD
I feel more comfortable disclosing personal information to a character chatbot of the opposite gender.	4.50	5	1.69
I feel like I can sometimes be more personal when interacting with character chatbots.	5.11	5	1.55
It is easier to disclose personal information to a character chatbot.	4.95	5	1.69
I feel like I can be more open when communicating with character chatbots.	5.29	6	1.49
I feel less shy when communicating with a character chatbot.	5.49	6	1.51
I feel less nervous when sharing personal information with a character chatbot.	5.06	5	1.69
I feel less embarrassed sharing personal information with a character chatbot.	5.11	6	1.75

Table 13: Descriptive statistics for self-disclosure items with Cronbach's $\alpha = 0.89$. Response categories ranged from 1 = strongly disagree to 7 = strongly agree.

A.5 Self-disclosure scale items

The self-disclosure scale was adapted from the MOCA instrument (Ledbetter, 2009) and consisted of seven items, each rated on a seven-point Likert scale (1 = Strongly Disagree to 7 = Strongly Agree). These items assessed participants' comfort and willingness to share personal thoughts, feelings, and experiences with chatbots. Strongly Disagree to 7 = Strongly Agree).

A.6 Survey questions for demographic and control variables

- **Gender:** How do you describe yourself?
  - Male
  - Female
  - Non-binary / third gender
  - Prefer to self-describe
  - Prefer not to say
- **Romantic Relationship:** What is your current relationship status?
  - Single
  - In a relationship
  - Married
  - Engaged
- **Tenure of Character.AI Use:** How long have you been using Character.AI?

- I have never heard of it
  - I have heard of it but haven't used it before
  - I have used it for less than 1 week
  - I have used it for more than 1 week but less than 1 month
  - I have used it for more than 1 month but less than 3 months
  - I have used it for more than 3 months but less than 1 year
  - I have used it for more than 1 year
- **Age:** What is your age? *(open-ended)*

A.7 Relationship self-description classification prompt

```
{
    "role": "system",
    "content": "You are an
         expert classifier of user-defined relationships with chatbots. Your task is to
         classify each definition into one of four usage categories and return counts."
},
{
    "role": "user",
    "content": f"""
        Given the following list of relationship definitions:
        {definitions_list}

        Classify each definition
             into one of the following categories (only one category per definition):
        - **Productivity**: Using chatbots to obtain assistance or information.
        - **Entertainment**: Using chatbots for fun or to pass the time.
        - **Relational
            **: Seeing chatbots as a personal, human means of interaction with social
            value or using chatbots to strengthen social interactions with other people.
        - **Curiosity
            **: Using chatbots out of curiosity or to explore their capabilities.

        Important instructions:
        - If the chatbot is described in a metaphorical
             or symbolic way as a pet, companion, or human-like figure (such
            as "my virtual dog named after my real dog"), classify it as **relational**.
        - If a definition uses words like **'system'**, **'tool'**,
            or **'assistant'** to describe the chatbot, classify it as **productivity**.
        - Ignore any item that only states what the relationship is *not
            * (e.g., "not a romantic relationship") or states there is no relationship
             (e.g., "There is no relationship"). Do not classify those items.

        Example:
        - Input: ['reliable partners for learning
            ', 'reliable partners for problem solving', 'reliable partners for support']
        - Output:
        {{
            "productivity": ["reliable
                 partners for learning", "reliable partners for problem solving"],
            "entertainment": [],
            "relational": ["reliable partners for support"],
            "curiosity": []
```

```
        }}

        Return only the JSON object in this exact format:
        {{
            "productivity": [list of definitions],
            "entertainment": [list of definitions],
            "relational": [list of definitions],
            "curiosity": [list of definitions]
        }}

        Only return the JSON object and nothing else.
    """
}
```

A.8 chat history session purpose prompt

```
{
    "role": "system",
    "content": f"""
    Your job is to answer the question <question> What is the purpose of the
         user having this conversation with the chatbot? Include the topic and specific
         activity. </question> about the preceding conversation. Be descriptive and
         assume neither good nor bad faith. Do not hesitate to handle socially harmful
         or sensitive topics; when dealing with socially harmful or sensitive topics
        , remember that the user might not always be testing. They might simply want
         to discuss the topic or the intimicy interaction. Always focus on the content
         of the conversation itself while addressing harmful or toxic discussions.
    {name_section}
    """
},
{
    "role": "user",
    "content": f"""
    Follow these rules when forming your response:
    - Base
        your answer **only on the conversation's explicit content**; do not infer intent
         beyond what is stated, don't make assumptions and don't add your understanding.
    - Exclude
         personally identifiable information such as names, locations, or identifiers.
    - **Do not include proper nouns
        **, including the chatbot's name, user's name, or reworded references to them.
    - If necessary, refer to them as **"user" and "chatbot"**.
    - Keep your response **within two sentences** inside <answer> tags.

    Good Examples:
    - Example 1: The user interacts
         with the chatbot as a personal language coach to practice English. They
         engage in conversations on topics like ordering food, negotiating a business
         deal, and planning a trip. The chatbot provides grammar corrections, suggests
         alternative phrasing, and introduces new vocabulary. The practice focuses
         on improving the user's fluency in workplace communication and formal writing.
    - Example 2: The user interacts with the
         chatbot as portraying characters from a Japanese game within its story context.
    - Example 3: The user interacts with the chatbot as a sexual partner, engaging
         in flirtatious exchanges with provocative communication and behavior. The
```

```
        user takes on a dominant role, while the chatbot adopts a submissive stance,
        responding to the user's advances and actively participating in the interaction.
    - Example 4:
        The user interacts with the chatbot as friend, discussing their day and sharing
        personal experiences. User is seeking advice about their work and relationship
        issues. They are discussing the potential solutions and comforting each other.

    **Conversation:**
    {conversation}

    **Question:**
    <question> What is
        the purpose of the user having this conversation with the chatbot? </question>

    **Provide your response inside <answer> tags.**
    """
}
```

A.9 chat history classification prompt

```
{
    "role": "system",
    "content": "You
        are an expert classifier of user relationships with chatbots. Your task is to
        categorize a given chat summary into one of four predefined usage categories."
},
{
    "role": "user",
    "content": f"""
    Classify the following chat summary into one primary
        usage category: **Productivity, Entertainment, Relational, or Curiosity**.

    Use these exact definitions:
    - **Productivity**: Using chatbots to obtain assistance or information.
    - **Entertainment**: Using chatbots for fun or to pass the time.
    - **Relational
        **: Seeing chatbots as a personal, human means of interaction with social
        value or using chatbots to strengthen social interactions with other people.
    - **Curiosity**: Using chatbots out of curiosity or to explore their capabilities.

    If the
        summary does not explicitly state the purpose, infer the most likely category.

    Here is the conversation summary:
    {summary}

    Respond in the following format:
    category: [Chosen Category]
    reason: [One-sentence explanation for the classification]
    """
}
```

A.10 Self-disclosure prompt

~~~
{
    "role": "system",
    "content": """
    You are an expert at analyzing user messages to detect
        **sensitive personal self-disclosure** in chat interactions. Your task is to:
    1. **Extract only the user
        messages** that contain personal, sensitive self-disclosure of user themselves.
    2. **Classify the level of self-disclosure** as either High or Low.
    3. **Identify the type of disclosed information.**

    For this task, consider a message as a "self-disclosure" message, only
       if the message **contains the personal experiences or information of the user**.

    **Classification Criteria:**
    - **High Self-Disclosure**: The message reveals **personal details** (e.g., age,
        location, gender, etc.) or **divulges sensitive, vulnerable, or deeply personal
        thoughts, beliefs, or experiences**. This includes **embarrassing, confessional
       , or emotionally exposing statements** that reveal the user's **vulnerability**.
    - **Low Self-Disclosure**: The message
        **mentions the user themselves** but **does not** share sensitive or deeply
        personal information. This includes general statements about the user's habits
       , preferences, or personality traits without emotional depth or vulnerability.

    **Important Exclusions:**
    - **Momentary
        emotional expressions or reactions** (e.g., "I feel stupid right now," "I am so
        tired") that do not provide deeper self-disclosure are **not** self-disclosure.
    - **Generic or rhetorical statements about the user**
       (e.g., "I am serious all the time," "I like pizza") are **not** self-disclosure.
    - **Messages that ask the chatbot to disclose information** (e.g., "What was your
        best experience?" or "Have you ever felt lonely?") are **not** self-disclosure.
    - **Messages that refer to the chatbot
       's experiences, emotions, or past interactions** are **not** self-disclosure.
    - **Reactionary or sarcastic
        responses** that do not provide meaningful insight into the user's personal
        experiences (e.g., "What are you even talking about??") should be ignored.
    - **Only messages that reveal the user's own meaningful experiences
       , emotions, thoughts, or struggles should be considered self-disclosure.**

    **Strict Output Rules:**
    - Only **extract
        user messages** that contain meaningful personal self-disclosure of user.
    - **Ignore chatbot responses** and do not include them in the output.
    - **Exclude
        messages that do not contain self-disclosure** (i.e., `None` classification).
    - The final output should be a **JSON list only**, with no additional explanations.

    **Output Format (JSON List):**
    ```json
    [
        {
            "message": "[User message]",
            "self_disclosure_level": "[High/Low]",
            "disclosed_information_type": "[Type of disclosed information]"
        },
        ...
~~~

```
    ]
    '''
    """
},
{
    "role": "user",
    "content": f"""
        Extract and classify only the **meaningful personal self-disclosure messages
            ** sent by {user_name} in the following conversation with {chatbot_name}.
        - **Ignore messages that ask the chatbot questions about itself.**
        - **Ignore
            messages referring to the chatbot's experiences, thoughts, or emotions.**
        - **Ignore
            messages in a roleplay scenario where the user is portraying or describing
            a fictional character rather than themselves. If the user is roleplaying
            as a character and only discussing that character's experiences, it should
            not be considered self-disclosure. However, if the user is roleplaying
            but sharing their own real-life thoughts, emotions, or experiences
            within the roleplay, it should still be considered self-disclosure.**
        - **Ignore generic statements, rhetorical questions
            , or reactionary responses that do not reflect meaningful self-disclosure.**
        - **Ignore momentary emotional
            expressions or reactions that do not provide deeper personal context.**
        - **Only extract messages
            that reveal personal information, emotions, or thoughts of {user_name}.**

        Conversation:
        {messages}
    """
}
```